

CURBING THE DRIFT: HOW REGULATED CONVERSATIONAL FLOWS SHAPE LLM PERSONA CONSISTENCY

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by

ALCANTARA, Van Asher
ENCARGUEZ, Jorenie
GAN, Austin Philippe
GUTIERREZ , Allen Andrei

Ethel ONG
Adviser

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We declare that this thesis proposal was completed in accordance with institutional policies on the responsible use of Generative Artificial Intelligence (AI). Any use of Generative AI tools served only to support human work and judgment. We retain full responsibility for the originality, correctness, ethical conduct, and academic integrity of this thesis.

AI-assisted outputs, where applicable, were critically reviewed, verified, revised, and integrated by us prior to inclusion.

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- **R** – Refinement
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Code generation or software development	✓			
Visualization or plotting code	✓			
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We used the following Generative AI tools:

- ChatGPT
- Gemini

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Abstract

Large language models (LLMs) have enabled more natural and engaging conversational systems, particularly in applications involving fictional characters and interactive storytelling. As conversations progress, models often exhibit persona drift, where tone, behavior, and identity gradually deviate from the intended character, reducing narrative coherence and user immersion. Maintaining a consistent persona across extended multi-turn dialogues remains a persistent challenge. While prior work has largely addressed this issue through model-level approaches, these methods are computationally intensive and do not fully account for the role of interaction design. This study investigates how interaction constraints, specifically dialogue flow and input constraints, affect persona consistency in LLM-driven dialogues. An interaction layer-based system is developed to regulate conversational flow without modifying the underlying model architecture, allowing the effects of interaction-level design to be isolated. The research follows a two-phase evaluation framework consisting of a baseline assessment using unconstrained dialogues and a controlled comparative evaluation across multiple system configurations. Performance is measured using quantitative metrics, including persona adherence rate and interaction entropy, alongside qualitative user evaluations of narrative coherence and immersion. The findings aim to determine whether interaction-level constraints can effectively reduce persona drift and improve long-term consistency in a scalable and cost-efficient manner, contributing to more reliable and immersive character-driven conversational systems.

Keywords: Large Language Models (LLMs); Persona Consistency; Persona Fidelity; Persona Drift; Multi-Turn Dialogue; Interaction Constraints; Dialogue Flow; Conversational AI; Narrative Coherence; Out-of-Persona (OOP).

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Chapter 1

Introduction

1.1 Background of the Study

Large Language Models (LLMs) have rapidly advanced, showing considerable promise in conversational AI applications. They can generate human-like, fluent dialogue and are increasingly used as virtual assistants in customer service (Shi & Niu, 2024; Schreiber, White, & Schmidt, 2024), as well as interactive storytelling and character-based chatbot systems (Shi & Niu, 2024; Bohdan, Tvoroshenko, Gorokhovatskyi, & Kobylín, 2025). The ability of these systems to mimic natural conversation is largely due to the underlying transformer architectures, which excel at capturing and generating coherent dialogue.

In conversational AI systems, a persona represents the set of behavioral traits, tone, and identity that guide how a language model communicates with users, enabling more coherent character behavior and improving user engagement in dialogue-based applications. LLMs often adopt a neutral persona, often friendly, helpful, and intelligent, which contributes to their ability to engage in seamless conversations with users (Schreiber et al., 2024). However, maintaining persona consistency over extended dialogues remains a challenge. As conversations progress across multiple turns, the model’s tone, voice, and behavioral characteristics may begin to drift over the course of a conversation, leading to inconsistencies in the character’s behavior.

The mechanism behind this degradation is rooted in the structure of the interaction itself. Each conversational turn introduces new contextual input that competes with the model’s original persona instructions for attention. As dialogue length increases, the transformer’s attention to the initial persona-defining tokens

progressively decays, a phenomenon documented by K. Li et al. (2024), causing the model to weigh recent conversational context more heavily than its assigned character. This effect is compounded when user inputs are unstructured: abrupt topic shifts, anachronistic references, or prompts that push the character outside its defined knowledge boundaries all introduce contextual pressure that pulls the model away from persona fidelity. As this pressure accumulates across turns, the model increasingly prioritizes surface-level response coherence over character consistency, causing the character’s voice and behavior to gradually revert toward the model’s generic defaults. This drift is not a sudden failure but a gradual erosion, worsening as dialogue length increases.

Emerging work on turn-taking dynamics and interaction design suggests these factors may influence dialogue quality broadly, though their specific effect on persona drift of fictional characters remains untested. There are several approaches to combating persona drift. Techniques like fine-tuning, Reinforcement Learning (RL), and Retrieval-Augmented Generation (RAG) have been proposed to improve persona consistency. Huang et al. (2025) explored reinforcement fine-tuning to improve character fidelity in chatbots, while Shi and Niu (2024) proposed counterfactual datasets to fine-tune LLMs for enhanced persona consistency. Ji et al. (2025) introduced contrastive self-play alignment to fine-tune LLMs for better role-playing behavior, demonstrating the value of targeted persona training.

While these model-level solutions show promise, they often come at a high computational cost or require expert intervention. Findings from recent studies suggest that interaction-level orchestration can influence model consistency, potentially serving as a mechanism to prevent persona misalignment without modifying the underlying model while offering a more potentially scalable solution. Sharma et al. (2026) argue that Interface Scaffolds, such as structured input prompts or feedback forms, help guide user input, improving model performance. Similarly, Tam et al. (2024) and Mündler, He, Jenko, and Vechev (2024) discuss the impact of constraints like turn limits and format restrictions on LLM performance.

Despite these insights, there remains limited empirical investigation into how interaction-level constraints directly affect persona consistency in extended multi-turn dialogues. Most existing studies focus on improving model behavior through internal modifications, while the potential of external interaction control mechanisms remains underexplored. In particular, there is a lack of systematic evaluation comparing unconstrained and constrained interaction settings using both quantitative metrics and user-based assessments. Addressing this gap is essential to determine whether interaction-level interventions can effectively reduce persona drift, stabilize persona behavior, and maintain narrative coherence without incurring the computational costs associated with model retraining.

The problem of persona drift in LLMs remains unsolved despite advancements in fine-tuning and memory management techniques. While these methods improve memory recall, they fail to address how interaction constraints embedded in the turn-taking dynamics can help maintain persona consistency in extended multi-turn dialogues. As a result, there remains a lack of empirical evidence regarding whether these interface-level constraints can effectively mitigate persona drift in long-form dialogue without the computational overhead of model retraining. Understanding how interaction constraints, such as turn-taking dynamics and structured input limitations, influence persona consistency and narrative coherence during extended multi-turn dialogues may mitigate persona drift and enhance user immersion.

1.2 Research Objectives

The aim of this research is to investigate how interaction constraints, specifically dialogue flow and input constraints, influence persona consistency and narrative coherence during extended multi-turn dialogues. To achieve this general objective, the study focuses on the following specific objectives:

1. Determine the interaction constraint strategies and baseline character parameters to establish a structured reference point for evaluating persona consistency across extended multi-turn dialogues.
2. Develop the knowledge base and interaction layer that will implement the character persona and interaction constraints between the LLM and user.
3. Evaluate the performance of the interaction layer to determine whether interaction-level constraints effectively reduce persona drift.

1.3 Scope and Limitations of the Research

For our first objective, we narrowed the constraints to two categories. Input Constraints evaluate the user’s message before it reaches the model, checking out-of-persona compatibility (Barrot, 2026), topic relevance (Barrot, 2026; Zhang et al., 2023), historical consistency (Khan et al., 2025; Zhang et al., 2023), and guided input compliance (Zhang et al., 2023; Barrot, 2026). Dialogue Flow Constraints regulate how the conversation progresses once accepted by tracking conversation state (Khan et al., 2025; Zhang et al., 2023), narrative progression (Zhang et

al., 2023), context continuity (Khan et al., 2025; Zhang et al., 2023), and topic boundaries (Zhang et al., 2023; Barrot, 2026).

For this study, Sherlock Holmes, the fictional detective created by Sir Arthur Conan Doyle, was selected as the target role-playing character. Sherlock Holmes was chosen because his personality, investigative methods, communication style, moral principles, and knowledge boundaries are extensively documented across multiple canonical literary works, making him an appropriate subject for evaluating persona consistency in multi-turn conversations. Baseline character parameters are therefore derived from these publicly available literary works, with behavioral traits, linguistic patterns, and narrative identity encoded through structured prompt-based instructions. Accordingly, Sherlock Holmes is the only supported persona in this study, as the knowledge base is constructed exclusively from his canonical literary works. Support for additional fictional characters is outside the scope of the present research. The use of a single, well-documented fictional character with traceable linguistic patterns and established knowledge boundaries provides a standardized reference for evaluating persona consistency across all experimental conditions (N. Wang et al., 2024; Ran et al., 2024). Prompt-based persona encoding remains sensitive to variations in wording and contextual framing, which may introduce behavioral variability across extended interactions (Chang et al., 2025). The scope is also limited to a static persona, deliberately excluding narratively motivated character transformations such as plot-driven personality shifts which the current adherence metrics would incorrectly penalize (Costa, Alves, & Vicente, 2026). Finally, the character’s specific cultural and historical origins may limit the generalizability of findings to characters from other cultural contexts (Chang et al., 2025).

The scope of the proposed chatbot is confined strictly to fictional, persona-driven role-play. Every interaction is framed within a Sherlock Holmes investigative scenario, and the system is not designed to sustain open-domain conversation or answer queries unrelated to the character’s persona or the active case. Interactions that fall outside this fictional, investigative frame are treated by the middleware as out-of-scope, regardless of their content, consistent with the narrative-output design of the framework discussed later in this section.

Consistent with this scope, participant evaluation is built around a role-playing dialogue rather than an open question-and-answer exchange. During each testing session, participants take on the role of a client or associate presenting a fictional case to Sherlock Holmes, while the chatbot, embodying Holmes’ persona, gathers details, asks probing questions, and works through the investigation using his characteristic deductive method. This mirrors the canonical dynamic between Holmes and his interlocutors (e.g., a client or Dr. Watson) found in Doyle’s original works, giving participants a familiar narrative frame for judging persona and narrative

consistency.

Designing and developing a custom interaction constraint system involves creating dialogue prototypes and an interaction layer that regulates conversational flow through fixed turn limits, structured turn-taking, and restricted input formats. The use of a separate interaction layer, rather than modifying the underlying model, enables the isolation and direct evaluation of interaction-level constraints independent of architectural changes. This approach is supported by Tam et al. (2024), who demonstrate that controlling interaction formats impacts LLM performance, and by Schreiber et al. (2024), who emphasize the importance of structured mechanisms in maintaining contextual consistency. To ensure legal compliance and a stable narrative baseline, development is limited to publicly available fictional characters. However, this approach has inherent limitations. First, since such frameworks are designed for narrative outputs (N. Wang et al., 2024), the system is not applicable to non-fictional or task-oriented conversational agents. Second, the system remains constrained by the finite context window of the underlying LLM. While the interaction layer possibly reduces persona drift, it cannot extend memory capacity or exceed token limits, as maintaining contextual equilibrium remains a fundamental challenge (Dongre et al., 2025).

The proposed framework employs two open-weight large language models with distinct responsibilities to separate response generation from response evaluation. Gemma-3 12B Instruct is used as the primary response generation model because it provides a balance between conversational quality, computational efficiency, and local deployability, making it suitable for evaluating interaction-layer middleware without relying on proprietary cloud-based models. Llama 3.1 8B Instruct is utilized as the Judge LLM for the Input Constraint Engine and Output Validation Engine due to its strong instruction-following capability and consistent performance in reasoning and evaluation tasks. By assigning separate models for generation and evaluation, the framework minimizes self-evaluation bias and enables independent assessment of user inputs and generated responses. However, the findings remain bounded by the capabilities and limitations of the selected open-weight models. Variations in reasoning ability, instruction adherence, and response quality across other LLMs may produce different results; therefore, the conclusions of this study should be interpreted within the context of the chosen models rather than generalized to all large language models.

Evaluation of the system’s performance combines both automated and manual approaches. Automated evaluation analyzes backend dialogue logs, including turn counts, input lengths, and contextual shifts, to compute persona adherence rate and interaction entropy reduction, where persona adherence rate measures the proportion of responses that remain consistent with the defined character’s traits, tone, and identity, and interaction entropy reduction quantifies the decrease

in variability and unpredictability of dialogue flow under constrained conditions, following methods from N. Wang et al. (2024) and Ji et al. (2025). Manual evaluation complements these metrics through user ratings and qualitative assessments, providing ground truth for subjective aspects such as narrative coherence and character consistency (N. Wang et al., 2024). Manual evaluation complements these metrics and serves as the primary means of assessing subjective, qualitative aspects of the system such as narrative coherence and character consistency, since these dimensions are difficult to fully capture through automated scoring alone (Wang et al., 2024). This qualitative assessment is carried out through participant-based evaluation rather than expert literary review: undergraduate participants drawn from any academic field or discipline, provided they possess sufficient familiarity with Sherlock Holmes’ character to recognize deviations in persona and narrative flow, rate each configuration after their role-playing session. These user ratings serve as the qualitative ground truth for narrative coherence and immersion, and are analyzed alongside the automated persona adherence and interaction entropy metrics to provide evidence on whether interaction-level constraints reduce persona drift. However, the evaluation is limited to fictional character systems and excludes real-world applications. Additionally, results depend on the initial prompt design, as baseline persona stability can influence adherence outcomes (Chang et al., 2025). Finally, since the approach applies external interaction constraints without modifying the underlying model, findings reflect the impact of input-level control rather than the full capabilities of internally adapted models.

1.4 Methodology

The study adopts a design and development research approach to develop and evaluate a persona-consistent role-playing chatbot based on the fictional character Sherlock Holmes. The research is carried out through five major phases: Persona Knowledge Base Construction and Validation, Interaction-Layer Middleware Development, Framework Validation, System Testing and Evaluation, and Documentation.

Phase 1: Persona Knowledge Base Construction and Validation

The formal execution of the first phase is scheduled over a 5-week period spanning from the third week of September to the second week of October. It first phase focuses on constructing the Persona Knowledge Base (PKB) that serves as the knowledge foundation of the chatbot. Sherlock Holmes is selected as the target role-playing character because his personality, investigative methods, com-

munication style, beliefs, moral principles, and knowledge boundaries are extensively documented across canonical literary works, providing a stable reference for persona construction.

Relevant literary sources are collected and analyzed to extract persona-defining information. The extracted characteristics are organized into structured knowledge components that represent Sherlock Holmes’ personality, expression, knowledge, moral principles, and beliefs. These components form the Persona Knowledge Base that guides the chatbot’s responses throughout the conversation.

Since the study utilizes existing literary works, only publicly available and public-domain Sherlock Holmes texts are used. All literary sources are properly cited and utilized solely for academic and non-commercial research to ensure the ethical use of intellectual property. Upon completion, the constructed Persona Knowledge Base is reviewed and validated to verify that it accurately represents the intended Sherlock Holmes persona before being integrated into the proposed framework .

Following a 2-week construction period, the Persona Knowledge Base is reviewed and refined over the subsequent 3 weeks to verify that it accurately represents the intended persona before being integrated into the proposed framework.

Phase 2: Interaction-Layer Middleware Development

The second phase involves the development of the proposed interaction-layer middleware that manages the overall conversational process. Using the validated Persona Knowledge Base as its foundation, the middleware is developed to regulate interactions before and after response generation. The framework consists of the Input Constraint Engine, Decision Layer, Prompt Builder, Dialogue Flow Manager, Response Generation Layer, and Output Validation Engine. These components collectively evaluate user inputs, maintain dialogue continuity, construct prompts, generate responses, and validate generated outputs to improve persona consistency during extended conversations. This phase runs for a total duration of 9 weeks, beginning concurrently with Phase 1 in the third week of September and concluding by the third week of November.

The middleware is integrated with the Persona Knowledge Base to form the complete conversational framework. The language models, software libraries, and development tools utilized throughout this phase are used in accordance with their respective licenses and terms of use, and the developed system is intended solely for academic research. It is progressively implemented, moving from core architecture in September to chat interface integration in October and finalizing with logging modules in November to form the complete conversational framework, en-

suming all language models, software libraries, and development tools are used in accordance with their respective licenses.

Phase 3: Framework Validation

The third phase focuses on validating the completed framework before it is evaluated with participants. This phase is allocated a concise 2-week duration spanning the fourth week of November to the first week of December. Functional and integration testing are conducted to ensure that the Persona Knowledge Base, Input Constraint Engine, Decision Layer, Prompt Builder, Dialogue Flow Manager, Response Generation Layer, and Output Validation Engine operate as intended.

The study prepares four system configurations for comparative evaluation:

1. Baseline persona prompting
2. Baseline with Input Constraint Engine and Decision Layer
3. Baseline with Dialogue Flow Manager
4. The complete WATSON framework

Each configuration is tested using the same predefined dialogue scenarios and evaluation conditions. Automated evaluation measures are used to assess persona consistency, dialogue coherence, historical consistency, style adherence, response latency, and system performance. This phase also verifies that the four configurations are stable and ready for participant-based evaluation. Any technical issues identified during testing are resolved before proceeding to the next phase.

Phase 4: System Testing and Evaluation

Following technical validation, the four system configurations will undergo participant-based evaluation to determine their effectiveness in maintaining Sherlock Holmes’ persona during extended conversations. It will be conducted over a 5-week period spanning from the third week of January to the third week of February. A total of 20 undergraduate participants, aged 18 years or older and possessing sufficient familiarity with Sherlock Holmes’ personality, communication style, and investigative behavior, will be recruited through convenience sampling.

Prior to participation, each participant will receive an informed consent form describing the purpose of the study, evaluation procedures, expected duration, data to be collected, potential risks, and participant rights, including the right to withdraw from the study at any time without penalty. Participation in the study is entirely voluntary. The study presents minimal risk because participants interact only with an AI chatbot portraying the fictional character Sherlock Holmes, and

no sensitive personal information or psychologically distressing topics are intentionally introduced as part of the evaluation. The chatbot will also disclose that it is an artificial intelligence system portraying the fictional character Sherlock Holmes.

Each participant will evaluate all four system configurations using a within-subjects experimental design. To minimize fatigue, learning effects, and familiarity bias, each configuration will be evaluated in a separate testing session, with one configuration assigned per session. Consequently, each participant will complete four independent testing sessions, corresponding to the following configurations:

1. Baseline persona prompting
2. Baseline with Input Constraint Engine and Decision Layer
3. Baseline with Dialogue Flow Manager
4. The complete WATSON framework

During each testing session, participants will interact with the assigned configuration through a multi-turn role-playing conversation based on the same or equivalent fictional investigative scenario. The order in which participants evaluate the four configurations will be randomized to minimize ordering effects and ensure a fair comparison.

Upon completing each testing session, participants will complete evaluation questionnaires assessing the System Usability Scale (SUS), perceived persona consistency, and perceived response wait time. Conversation logs generated during each interaction will also be collected and anonymized for analysis. No personally identifiable information beyond what is necessary for participation will be collected, and all research data will be stored in password-protected storage accessible only to the researchers.

The participant responses will be compared across the four testing sessions to determine how each interaction-layer mechanism influences perceived usability, responsiveness, and persona consistency. These findings will be analyzed together with the automated evaluation metrics obtained during the framework validation phase to provide a comprehensive assessment of the proposed WATSON framework.

Phase 5: Documentation

The final phase consists of analyzing the results obtained from framework validation and participant-based evaluation. While report writing is scheduled continuously across the entire 12-month project timeline (May to April), the dedicated

analytical portion of the final phase spans a 6-week period from the fourth week of February through the end of March. Results from the automated evaluation metrics and participant questionnaires are summarized and compared across the evaluated system configurations, while observations from participant feedback and conversation logs are analyzed to identify recurring patterns related to persona consistency and conversational coherence.

The findings are interpreted to determine whether the proposed interaction-layer middleware effectively reduces persona drift while maintaining a consistent Sherlock Holmes persona throughout extended conversations.

Academic Year		2025-2026				2026-2027							
Thesis Subject		THS-ST1				THS-ST2				THS-ST3			
Month		MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR
of Weeks		4	4	5	4	5	4	4	5	4	4	5	4
	Fictional character selection and data sourcing	o o X o											
	Identification of Baseline Persona Parameters	o o o X	X X o o										
	Identification of Interaction Constraint Strategies	o o o X	X X o o										
	Persona construction pipeline design		o o X X	X X o o o									
	Middleware architecture design		o o X X	X X o o o									
Phase 1	Persona knowledge base construction					o o X X o							
	Persona profile validation and refinement					o o o o X	X X o o						
	Interaction-layer middleware implementation					o o X X X	X X X X						
Phase 2	Chat interface and interaction constraint implementation						o o X X	X X o o					
	Conversation logging and evaluation module implementation							X X X o					
Phase 3	Pilot testing and system refinement							o o o X	X o o o				
Phase 4	Participant recruitment									o o X X	X X X o		
	Experimental evaluation (Baseline and Interaction-Constrained Systems)									o o X X	X X X o		
Phase 5	Data analysis and comparative benchmarking										o o o X	X X o o o	
	Final Revisions											o o X X X	
	Documentation and Report Writing	XXXX	XXXX	XXXXX	XXXX	XXXXX	XXXX	XXXX	XXXXX	XXXX	XXXX	XXXXX	XXXX

Table 1.1: Gantt Chart

1.5 Significance of the Research

The study contributes to advancements in Conversational AI, Human-Computer Interaction, and interactive storytelling. It provides an interaction-level constraint layer, through structured turn-taking and guided input mechanisms, to improve persona consistency in character-based chatbot systems. By measuring persona drift using metrics such as persona adherence rate, interaction entropy, and user ratings of coherence and persona consistency, the study introduces a replicable framework for evaluating and maintaining persona consistency.

The findings of this research are particularly significant for AI development teams working on character-based chatbot systems. AI developers gain actionable strategies for implementing structured turn-taking as a cost-efficient alternative or complement to model retraining, enabling more predictable persona behavior. Machine learning engineers benefit from insights into how conversational structure affects model behavior, allowing them to integrate a lightweight interaction layer to enforce structured turn-taking, reduce inconsistencies, and enhance persona stability in a scalable manner, helping users remain immersed in the story and enhancing overall storytelling quality.

Beyond immediate development benefits, this research promotes more immersive and engaging interactions with AI characters. By stabilizing persona and maintaining narrative coherence, the findings aim to enhance users' sense of presence and continuity in long-form dialogues. While the primary application is interactive storytelling, the insights may also extend to other domains that require a stable persona.

Chapter 2

Related Works

In this section, we review relevant literature across three interconnected areas: first, role-playing conversational LLMs; second, persona drift, its manifestations, and current mitigation strategies; and third, interaction-level constraints.

2.1 Role-Playing Conversational LLMs

Role-playing language agents (RPLAs) represent a significant evolution from traditional question-answering systems, shifting the focus toward cognitive simulation that prioritizes personality consistency, behavioral rationality, and immersive dialogue (Y. Wang, Chen, & Xiao, 2026). To achieve this high level of character fidelity, recent literature has heavily focused on deepening the data representation within the models, whether through psychologically grounded personality scales, timeline-aware narrative extraction, emotionally annotated dialogue, constraint-optimization frameworks, or multi-turn reinforcement learning. Table 2.1 provides a comparative summary of these studies, outlining how each constructs its training data, sources its character or persona information, represents personality, and reports its limitations and key findings.

Table 2.1: Role-Playing Conversational LLMs

Keyword: Role- Playing Conversa- tional LLMs	Training Data Con- struction	Character /Persona Source	Persona- lity Repre- sentation Method	Limitation Ac- knowl- edged	Key Findings
(Capturing minds, not just words: Enhancing role-playing language models with personality-indicative data, Ran et al., 2024)	Gets responses from RPAs using a psychological scale questions Filters data using human-annotated personality labels as ground truth	46 characters from ChatHaruhi and RoleLLM (fictional or based from media)	Personality captured using psychological scale dimensions (OCEAN, MBTI, etc.)	Dataset is entirely AI generated which may have biases No human evaluation of outputs	Fine-tuning with personality-indicative data significantly improves character fidelity and "mind" capture compared to knowledge-only datasets.

(Living the novel: A system for generating self-training timeline-aware conversational agents from novels, Huang et al., 2025)	Teacher LLM generates preference pairs for character profiles Uses AI generated preference for GRPO fine-tuning	Characters extracted from a novel (one of Jules Verne’s books)	Multi-layered character profile Attributes, intrinsic drives, relationship dynamics Based on the current part of the timeline	Evaluated only one novel Did not test non-linear narratives	Character-first self-training is crucial for believability, while diegetic time constraints are crucial to prevent spoilers and maintain immersion.
(RoleCraft-GLM: Advancing personalized role-playing in large language models, Tao et al., 2024)	Annotated emotion real-world dialogues (social media, scripts, customer service)	20 custom-designed non-celebrity everyday Chinese personas (possibly office workers, students, etc.)	Emotion labels per utterance based on Ekman’s theory Profile has age, hobbies, and relationships	Only covers 20 Chinese personas which limits generalizability	Shifting focus to non-celebrity personas with emotional annotations improves realism and user engagement through nuanced dialogues.

(Optimizing Large Language Models for Dynamic Constraints through Human-in-the-Loop Discriminators, Wei & Miin, 2024)	Framework (CTOp) that extracts constraints from system interfaces and uses human/LLM discriminators to identify critical cases	A travel planner agent initialized by establishing constraints from evaluation interfaces	Concept optimization approach where learned rules are refined when exception cases arise	Common practices like model fine-tuning and reflection-based reasoning often address issues case-by-case which limits generalizability	Collaborating with human experts to identify critical cases allows LLMs to more effectively handle dynamic and complex application constraints.
(Consistently Simulating Human Personas with Multi-Turn Reinforcement Learning, Abdulhai et al., 2025)	Multi-turn reinforcement learning using consistency metrics (prompt-to-line, Q&A) as reward signals	A patient seeking counseling, a student, and an open-ended social chat partner	Consistency is tracked by checking how well responses match prompts, catching contradictions, and testing whether beliefs remain stable over time	While the framework offers a defined view of identity, it does not yet fully reflect how human behaviors change and develop over time	Using multi-turn RL with automated consistency rewards reduces persona drift by over 55%, resulting in more trustworthy simulated users.

Ran et al. (2024) elicit responses from role-playing agents using psychological scale questions and filter them using human-annotated personality labels, drawing

on 46 fictional characters from ChatHaruhi and RoleLLM. Personality is represented using the OCEAN and MBTI frameworks, capturing traits in a structured, quantifiable form rather than through unstructured dialogue alone. The authors note their dataset is entirely AI-generated, which may introduce bias, and that no human evaluation of outputs was conducted. Their key finding shows that fine-tuning with personality-indicative data significantly improves character fidelity and captures a character’s underlying ”mind” compared to knowledge-only datasets.

Huang et al. (2025) extract characters directly from a full-length novel, using a teacher LLM to generate preference pairs for each character profile, which are then used to fine-tune the model through Group Relative Policy Optimization (GRPO). Personality is represented through a multi-layered profile encoding attributes, intrinsic drives, and relationship dynamics, with character knowledge scoped to the current point in the story’s timeline. The authors acknowledge their evaluation was limited to a single novel and did not test non-linear narratives. Their key finding indicates that character-first self-training is crucial for believability, while diegetic time constraints are essential for preventing spoilers and maintaining immersion.

Tao, Xuechen, Shi, Yu, and Xie (2024), in their RoleCraft-GLM framework, use annotated emotional dialogues from real-world sources such as social media, scripts, and customer service transcripts to construct 20 custom-designed, non-celebrity Chinese personas. Personality is represented through emotion labels based on Ekman’s theory, combined with a profile of age, hobbies, and relationships. The authors note that covering only 20 Chinese personas limits generalizability. Their key finding shows that shifting focus to non-celebrity personas with emotional annotations improves realism and user engagement through nuanced and emotionally grounded dialogue.

Wei, Miin, and Miin (2024) propose the Constraint Optimization (CTOp) framework, which extracts behavioral constraints from system interfaces and uses human and LLM discriminators to identify critical edge cases, demonstrated through a travel planner agent. Rather than encoding personality, their approach centers on refining learned behavioral rules whenever exception cases arise. The authors observe that common practices like fine-tuning and reflection-based reasoning address such cases individually, limiting generalizability. Their key finding suggests that collaborating with human experts to identify critical cases allows LLMs to handle dynamic and complex constraints more effectively than native reasoning only.

Abdulhai et al. (2025) use multi-turn reinforcement learning with consistency metrics as reward signals, evaluated across three simulated personas: a counseling

patient, a student, and an open-ended social chat partner. Consistency is tracked through prompt-to-line and question-and-answer checks that catch contradictions and test whether beliefs remain stable over time. The authors acknowledge their framework offers only a fixed view of identity and does not reflect how human behaviors naturally change. Their key finding is that multi-turn RL with automated consistency rewards reduces persona drift by over 55%, producing more trustworthy simulated personas.

2.2 Persona Drift in LLMs

Persona drift remains a critical vulnerability in extended LLM interactions. Current literature identifies two primary vectors for this degradation: internal knowledge conflicts and external conversational entropy. Internally, conflicts between an LLM’s pretrained generalized knowledge and its assigned fictional context often cause the model to default to generic, uncharacterized outputs over time (Ji et al., 2025; Shi & Niu, 2024). Externally, OOP queries introduce entropy that forces inconsistent responses (Liu et al., 2022). Foundational to both vectors, however, is a mechanistic cause identified by K. Li et al. (2024). As conversations extend, the transformer’s attention allocation to system prompt tokens diminishes turn by turn, progressively eroding the influence of the original persona instruction and expanding the model’s output space toward generic defaults. K. Li et al. (2024) further demonstrate a particularly concerning side effect that the model can gradually begin adopting the conversational partner’s persona rather than its own. Choi, Hong, Kim, and Kim (2024) supports this pattern empirically across nine LLMs, finding that larger models are most vulnerable because their richer generative capacity leads them to fabricate self-referential details that compound into drift.

Table 2.2: Summary of Studies on Persona Drift in LLMs

Keyword: Persona Drift	Causes	Proposed Solution	Evaluation Method	Limitations
(Enhancing Persona Consistency, Shi & Niu, 2024)	Hallucination due to knowledge gaps Truthfulness shifts due to alignment bias Default assistant tone overriding character style	Provided a pipeline to generate persona- aligned dialogue datasets, augment with coun- terfactuals, filter for consistency, and fine-tuning	Consistency Score across prompts Compared with baseline models	Not tested in long multi-turn dialogue No real-time persona re- inforcement
(Persona Extending, Liu, 2022)	OOP queries cause fabricated or conflicting personas Long-tail bias leads to irrelevant persona selection	Provided a pipeline to retrieve compatible personas, verify with NLI, and generate responses with extended persona context	Consistency (ESIM), Quality (BLEU, ROUGE, CIDEr) Human Evaluation: relevance, entailment, fluency	Dependent on a predefined global persona set Cannot handle OOP queries outside this global persona set

(Persona-aware Contrastive Learning, Ji, 2025)	Weak persona grounding leads to generic outputs Pretrained knowledge conflicts with fictional context	Provided a pipeline to generate persona vs non-persona outputs, apply contrastive learning, and guide self-reflection for consistency	CA, CC, RA metrics (fluency, consistency, persona behavior) Human evaluation	Excessive reflection may reduce diversity Not validated in live multi-turn human interaction
(Beyond Single-Turn: A Survey on Multi-Turn Interactions, Li et al., 2026)	Inability to maintain coherence and instruction adherence as conversational length scales	Comprehensive survey analyzing instruction following, identifying external memory and structural integration as vital alternatives to model-centric fixes	Meta-analysis of multi-turn benchmarks, datasets, and instruction-following metrics	Highlights that standard prompt engineering fundamentally fails in prolonged interactions

(Moral Susceptibility and Robustness under Persona Role-Play, Costa et al., 2026)	Models lose assigned moral constraints over time and drift back to their default, built-in alignments	Introduced a strict mathematical benchmark to quantify "moral susceptibility" and "moral robustness" across different LLM families during persona role-play	Moral Foundations Questionnaire (MFQ) scoring	Focuses specifically on moral foundations rather than overarching behavioral traits
(Measuring and Controlling Instruction (In)Stability in Language Model Dialogs, Li et al., 2024)	Attention decay as dialogue length increases, transformer attention to system prompt tokens progressively weakens, allowing the model to drift toward generic behavior and even adopt the interlocutor's persona	Proposed split-softmax (SS), a training-free and parameter-free inference-time method that amplifies the model's attention weight allocated to system prompt tokens	Quantitative benchmark of 100 system prompts across 5 categories; stability measured via deterministic Python functions over self-chats between two instructed chatbots	All mitigations operate at the model or inference level and cannot eliminate drift entirely; Does not address interaction-level or interface-level constraints as a mechanism

(Examining Identity Drift in Conversations of LLM Agents, Choi et al., 2025)	Larger models introduce fictitious self-referential details during conversation, which are incorporated as credible identity information, causing compounding drift	No mitigation proposed; diagnostic study identifying parameter size, model family, and persona type as structural factors	BERTopic qualitative topic modeling; PsychoBench; Friedman test with Bonferroni-corrected Wilcoxon post-hoc tests across 3 conversational snapshots	No mitigation strategies proposed; Simplified personas may not reflect real-world complexity
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Shi and Niu (2024) identify hallucinations caused by knowledge gaps, alignment bias, and the tendency of language models to revert to their default assistant tone as major contributors to persona drift. To address these issues, they propose a pipeline that generates persona-aligned dialogue datasets, augments them using counterfactual examples, filters inconsistent samples, and fine-tunes the language model. Their approach evaluates persona consistency by comparing generated responses against baseline models using consistency scores across multiple prompts. Although the proposed pipeline improves persona adherence, the study evaluates only isolated prompts and does not investigate long multi-turn conversations. Furthermore, it does not provide mechanisms for reinforcing persona consistency during real-time interactions.

Liu et al. (2022) examines persona drift in multi-persona dialogue systems, where out-of-persona (OOP) queries may lead language models to fabricate conflicting identities or retrieve irrelevant persona information. The proposed Persona Extending framework retrieves compatible personas from a predefined persona repository, verifies their compatibility using Natural Language Inference (NLI), and incorporates the retrieved persona into response generation. Performance is evaluated using automatic metrics including ESIM, BLEU, ROUGE, and CIDEr, together with human evaluations measuring relevance, entailment, and fluency. However, the approach depends heavily on the availability of predefined personas and cannot effectively handle OOP queries that fall outside the existing persona repository.

Ji et al. (2025) attributes persona drift to weak persona grounding and conflicts between the pretrained knowledge of language models and their assigned fictional identities. The proposed Persona-aware Contrastive Learning framework generates persona-consistent and persona-inconsistent responses, applies contrastive learning to distinguish between them, and incorporates self-reflection mechanisms to improve consistency. The framework is evaluated using Character Accuracy (CA), Character Consistency (CC), Role Accuracy (RA), and human evaluation. Although the proposed method improves persona consistency, the study reports that excessive self-reflection may reduce response diversity and does not validate the approach in live multi-turn interactions involving human users.

Y. Li et al. (2026) provide a comprehensive survey of multi-turn interaction challenges in large language models. Rather than proposing a specific mitigation technique, the study analyzes existing instruction-following methods, benchmarks, and datasets to identify the major factors affecting long-term conversational coherence. The survey concludes that conventional prompt engineering alone is insufficient for maintaining instruction adherence as conversations become longer and highlights external memory mechanisms and structured interaction management as promising alternatives. Since the study serves as a survey, it does not experimentally evaluate a specific solution but instead synthesizes findings across existing literature.

Costa et al. (2026) investigate persona drift from the perspective of moral consistency. The authors observe that language models gradually abandon assigned moral characteristics during extended interactions and instead revert to their default alignment learned during pretraining. To quantify this phenomenon, they introduce mathematical benchmarks for measuring moral susceptibility and moral robustness using the Moral Foundations Questionnaire (MFQ). Although the benchmark provides a systematic way of evaluating moral consistency, its scope is limited to moral reasoning and does not evaluate broader behavioral characteristics that collectively define fictional personas.

K. Li et al. (2024) identify instruction instability as one of the fundamental causes of persona drift. Their analysis demonstrates that as conversations become longer, transformer attention gradually shifts away from the original system prompt, weakening persona instructions and allowing the model to generate increasingly generic responses or even adopt the conversational partner’s identity. To mitigate this behavior, they propose Split-Softmax (SS), a training-free inference-time technique that reallocates additional attention toward system prompt tokens. The approach is evaluated using a benchmark consisting of one hundred system prompts across multiple instruction categories with deterministic evaluation functions. Although Split-Softmax improves instruction stability, the authors acknowledge that inference-time attention modification alone cannot com-

pletely eliminate persona drift and does not address interaction-level constraints.

Choi et al. (2024) examine identity drift in conversational LLM agents by analyzing how generated self-referential information accumulates throughout multi-turn interactions. Their findings indicate that larger language models are particularly susceptible to introducing fabricated personal details, which subsequently become incorporated into the model’s perceived identity and compound over time. The study evaluates identity drift using BERTopic qualitative topic modeling, PsychoBench, and statistical analyses across multiple conversational snapshots. While the work provides valuable insights into the mechanisms underlying identity drift, it focuses primarily on diagnosis and does not propose methods for preventing or mitigating the observed behavior.

Collectively, these studies demonstrate that persona drift remains a significant challenge for role-playing language models, particularly during extended multi-turn interactions. Existing mitigation strategies primarily focus on modifying the language model itself through fine-tuning, retrieval mechanisms, contrastive learning, inference-time attention control, or external memory. However, relatively few studies investigate interaction-level control as a means of preserving persona consistency. This research addresses that gap by introducing a middleware-based interaction layer that regulates user inputs and dialogue progression before response generation, complementing existing model-centric approaches.

2.3 Interaction-Level Constraints

Interaction-level constraints refer to the external controls placed around an LLM to guide how it handles user input, manages context, follows assigned roles, and produces responses. In the context of this study, these constraints are important because a single instruction such as “act like Sherlock Holmes” may not be enough to keep the chatbot consistent throughout a long conversation. Table 2.3 presents studies that show how LLM behavior can be guided through structured prompting, dialogue-state control, process-based prompting, input moderation, multi-turn monitoring, and layered guardrails. These studies generally agree that LLMs become more reliable when their responses are shaped by external mechanisms instead of being left entirely to the model’s own generation process.

Table 2.3: Studies on Interaction-Level Constraints

Keyword: Interaction- Level Constraints	Interaction- Level Constraint Focus	Method	Key Finding	Constraint Design Contribution
(Empowering GPT as a Processual Writer: Didactext-Guided Prompting Improves Knowledge Access, Iterative Revision, and Overall Textual Quality, Mateo-Girona et al., 2025)	Process-guided generation constraints	Didactext-guided prompting through knowledge access, planning, production, and revision phases	Guided prompting improves reasoning depth, iterative refinement, structure, evidence use, and overall output quality.	Shows that interaction constraints can organize generation into stages before producing the final response, rather than allowing one-step unrestricted output.
(Infrastructure-Sentinel: Policy-Enforced Guardrails for Secure MCP-driven Infrastructure Agents, Kumar et al., 2026)	Multi-layer policy constraints	Guardian LLM with input filtering, tool validation, runtime gating, and post-action auditing	LLM agents can be controlled more reliably through multiple enforcement layers instead of one generic safety filter.	Shows that interaction constraints should be layered across the system, including input evaluation, planning control, output validation, and post-response review.

(NeuroFilter: Activation-Based Guardrails for Privacy-Conscious LLM Agents, Das & Fioretto, 2026)	Multi-turn trajectory constraints	Activation-based input filtering and activation velocity	Unsafe interaction patterns may emerge gradually across multiple turns, making single-turn input or output checking insufficient.	Shows that interaction constraints should be layered across the system, including input evaluation, planning control, output validation, and post-response review.
(Refining Input Guardrails: Enhancing LLM-as-a-Judge Efficiency Through Chain-of-Thought Fine-Tuning and Alignment, Kazemi Rad et al., 2025)	Input moderation constraints	Fine-tuned LLM-as-a-Judge guardrails with Chain-of-Thought reasoning	Input guardrails can detect malicious, jailbreak, or policy-violating prompts before they reach the main conversational agent.	Shows that interaction constraints should be layered across the system, including input evaluation, planning control, output validation, and post-response review.

(SGP-TOD: Building Task Bots Effortlessly via Schema-Guided LLM Prompting, Zhang et al., 2023)	Schema-guided dialogue constraints	Dialogue state tracking, task schema, policy prompting, and response generation	LLM dialogue systems perform better when guided by task schemas, belief states, and policy structures rather than relying only on the model’s internal generation.	Shows that interaction constraints should include a dialogue management layer that tracks context and determines the appropriate response direction before generation.
(The CLARIFIES+ Framework: A Contextualized Prompting Approach for Generative Artificial Intelligence, Barrot, 2026)	Prompt-structure constraints	CLARIFIES+ framework using context, limitations, audience, role, intent, format, inputs, end product, style, and safeguards	Prompts can act as boundary-setting instruments that define what the AI may produce, for whom, and under what conditions.	Shows that interaction constraints should include a dialogue management layer that tracks context and determines the appropriate response direction before generation.

Mateo-Girona, Kloss, and Lillo-Fuentes (2025) show that prompting can be more effective when the model is guided through a process rather than asked to produce an answer immediately. Their Didactext-guided approach divides writing into stages such as knowledge access, planning, production, and revision. This is useful for the present framework because the Sherlock Holmes chatbot should not

respond directly from raw user input alone. It should first understand the user’s message, connect it to the current context, prepare the right prompt, and check whether the response still fits the character before presenting it.

Kumar et al. (2026) present a stronger form of control through a multi-layer guardrail system. Their InfrastructureSentinel framework checks user input, validates tool use, controls execution, and audits the outcome after the action is completed. Although their study focuses on infrastructure agents, the idea is still relevant because it shows that one filter is usually not enough for complex LLM systems. For this study, the same principle applies to the middleware design, where input evaluation, dialogue flow management, prompt construction, and response validation work together to reduce persona drift.

Das and Fioretto (2026) focus on how risks can develop across multiple turns of conversation. Their NeuroFilter system shows that problematic behavior may not always be obvious in a single prompt, because the user’s intent can slowly build across several exchanges. This is important for persona stability because persona drift may also happen gradually. A Sherlock Holmes chatbot may not immediately break character, but repeated off-topic or role-breaking prompts can slowly move the conversation away from the intended fictional identity.

Rad et al. (2025) focus on input moderation through LLM-as-a-Judge guardrails. Their study shows that a separate model can evaluate incoming prompts and identify malicious, jailbreak, or policy-violating input before it reaches the main conversational agent. In the proposed framework, this supports the need for an Input Constraint Engine. This component can help prevent unsafe, irrelevant, or role-breaking prompts from influencing the Sherlock Holmes chatbot’s final response.

Zhang, Peng, Li, Zhou, and Meng (2023) provide a dialogue-focused example of interaction-level constraints through schema-guided prompting. Their SGP-TOD framework uses dialogue state tracking, task schemas, belief states, policy prompting, and response generation as separate parts of the dialogue process. This shows that an LLM can produce better responses when it is given structured information about the current conversation state. For this study, this supports the Dialogue Flow Manager, which helps decide how the Sherlock Holmes chatbot should respond based on the current context of the interaction.

Barrot (2026) contributes to the idea of structured prompting through the CLARIFIES+ framework. This framework organizes prompts by including context, limitations, audience, role, intent, format, inputs, end product, style, and safeguards. This is useful for the present study because maintaining the Sherlock Holmes persona requires more than simply naming the character in the prompt.

The Prompt Builder must clearly define the role, tone, context, and boundaries of the response so that the chatbot remains consistent with the intended fictional identity.

Taken together, these studies show that interaction-level constraints can be applied through staged prompting, schema-guided dialogue control, structured prompt design, input filtering, multi-turn monitoring, and layered validation. However, most of these studies apply such constraints to task-oriented systems, writing assistance, privacy protection, jailbreak detection, or infrastructure security. This study extends these ideas by combining several constraint mechanisms into one interaction-layer middleware for a Sherlock Holmes role-playing chatbot. The goal is to test whether using multiple coordinated constraints can further reduce persona drift and help preserve Sherlock Holmes’ character consistency across extended dialogue.

Chapter 3

Theoretical and Methodological Framework

3.1 Persona Theory

Transitioning a literary character into a computational entity requires an ontological framework that breaks identity down into measurable and verifiable components. Relying on unstructured text for persona descriptions may result in superficial imitation and unpredictable information extraction. Theoretical models in natural language processing demonstrate that character information must be mapped into structured relational links to ensure reliable and context-aware attribute retrieval during dialogue (Zhu et al., 2024). Furthermore, organizing these relational links into strict, machine-readable object models, such as JavaScript Object Notation (JSON) schemas, enables the system to distinguish a character’s identity into measurable subsets. This establishes a robust empirical benchmark for evaluating role-playing agents. Following this structural framework, the methodology anchors the persona baseline in a five-layer object model that synthesizes established concepts from computational linguistics, generative architecture, and machine ethics.

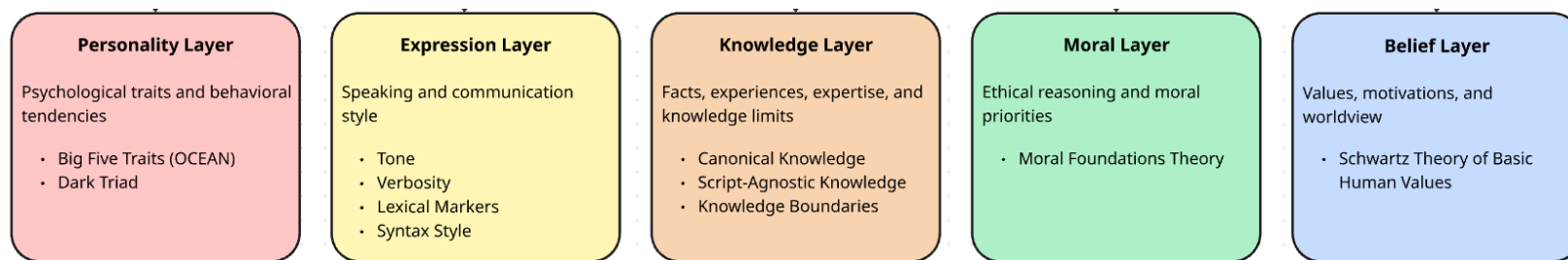


Figure 3.1: 5-Layer Persona Profile

3.1.1 The Personality Layer

The foundational layer of the persona relies on established trait psychology to dictate stable behavioral tendencies. As demonstrated by the InCharacter framework, role-playing agents must be anchored to quantifiable psychological scales to maintain consistency during interactions (X. Wang et al., 2024). Consequently, this layer explicitly adopts the Big Five Traits (OCEAN) model as its primary theoretical baseline, mapping the persona across five orthogonal dimensions. Furthermore, to capture complex, eccentric, or adversarial traits often present in literary figures, the layer integrates the Dark Triad model of personality (Lulla, Collins, Parekh, Hagendorff, & Kaplan, 2026). Together, these dual psychometric frameworks establish a standardized baseline to predict and evaluate character reactions.

3.1.2 The Expression Layer

A response may be factually and psychologically accurate while still failing to sound like the target persona. To address this, the Expression Layer is grounded in computational stylometry, specifically focusing on the theory of Speech Style Imitation. According to the RoleLLM framework, capturing a character’s distinct linguistic parameters is a necessary component of evaluating high-fidelity role-playing ability (N. Wang et al., 2024). Structuring expression as an independent layer ensures that the system measures and enforces specific behavioral attributes: tone, verbosity, lexical markers, and syntax style, preventing stylistic regression to the base model’s default tone.

3.1.3 The Knowledge Layer

In AI persona modeling, what a character cannot know is as theoretically important as what they do know. The Knowledge Layer is driven by the methodologies of role-specific knowledge constraints and memory injection utilized in advanced agent frameworks (N. Wang et al., 2024). This layer partitions information into three distinct dimensions: canonical knowledge, which refers to the core narrative facts of the character; script-agnostic knowledge, which refers to logical extensions of world facts; and strict knowledge boundaries, which define temporal, spatial, and technological limitations. This structure isolates memory streams and limits the LLM exclusively to the character’s canonical reality (Park, Park, & Lim, 2025), while establishing relational topologies that map dynamic interpersonal relationships (Zhu et al., 2024).

3.1.4 The Moral Layer

Evaluating a persona’s complex decision-making requires a structured framework for ethical reasoning. Research into LLM behavior demonstrates the phenomenon of “moral mimicry,” revealing that language models can and should be evaluated against specific human ethical structures to track behavior (Abdulhai et al., 2023). Therefore, the Moral Layer explicitly adopts Moral Foundations Theory (MFT) as its baseline. By evaluating the character’s internal logic against MFT’s distinct axes, such as (e.g., Care/Harm, Fairness/Cheating, and Loyalty/Betrayal), the system can mathematically assess whether the LLM’s generated advice aligns with the target persona’s established ethical code, rather than the default safety alignments of the base model.

3.1.5 The Belief Layer

While the Moral Layer dictates ethical rules, the Belief Layer governs intrinsic motivations. Recent paradigms in basic value alignment propose that evaluating an LLM’s underlying behavior requires mapping it against fundamental human value dimensions (Yao, Yi, Wang, Gong, & Xie, 2023). Consequently, this layer is anchored in the Schwartz Theory of Basic Human Values, which cross-culturally identifies foundational motivations such as self-direction, stimulation, security, and conformity. Structuring beliefs around the Schwartz values captures the character’s overarching worldview and biases, ensuring that the persona actively drives the dialogue in a direction consistent with their internal narrative motivations.

3.1.6 Persona Layer Interdependence

While defined as distinct computational vectors, these five layers do not operate in isolation. The theoretical strength of this object model lies in its interdependence, mirroring the cognitive architectures required for believable generative simulacra (Park et al., 2025). In this interconnected architecture, the Belief and Moral layers form the cognitive core, driving the decision-making logic, which is then filtered through the historical and relational constraints of the Knowledge Layer, and finally articulated through the stylistic parameters of the Expression and Personality Layers. By structuring the baseline persona as a highly interconnected, multi-dimensional relational matrix rather than isolated data points, the framework ensures that the evaluation rubric measures deep, holistic character fidelity (Zhu et al., 2024).

3.2 Conversation Management Theory

Conversation management theory provides the foundation for maintaining coherent and consistent interactions throughout multi-turn conversations. Unlike single-turn dialogue systems, conversational agents must continuously monitor the dialogue, maintain contextual information, and determine appropriate responses based on the current conversational state. Effective conversation management ensures that each response remains relevant to both the user’s current input and the overall objective of the conversation.

For role-playing conversational agents, conversation management is essential for preserving persona consistency during extended interactions. Without mechanisms that regulate the dialogue, large language models may gradually lose context, introduce contradictory information, or deviate from the intended character, resulting in persona drift (Budagam, 2024; Samuel et al., 2025). Consequently, conversation management advocates maintaining an explicit dialogue state that tracks conversational context and guides the progression of the interaction. These principles provide the theoretical basis for the Dialogue Flow Manager proposed in this study.

3.2.1 Dialogue State Tracking

Dialogue State Tracking (DST) maintains a structured representation of the current conversation by storing relevant contextual information such as user intent, dialogue history, and conversational goals. This allows conversational systems to generate responses that remain consistent across multiple dialogue turns. Recent studies have shown that explicitly maintaining dialogue state improves conversational coherence and reduces inconsistencies in large language models by allowing external management of conversational context rather than relying solely on the model’s internal memory (Khan et al., 2025; Zhang et al., 2023). These findings support the Dialogue State Tracking component of the proposed Dialogue Flow Manager, which monitors the progress of Sherlock Holmes’ investigation throughout the conversation.

3.2.2 Conversation Continuity

Conversation continuity refers to maintaining logical consistency throughout an interaction by ensuring that responses remain connected to previous exchanges. In long conversations, large language models often struggle to preserve contextual

information, causing them to forget earlier events or contradict previous responses (Ma et al., 2026). To address this limitation, conversation management emphasizes external context management rather than relying entirely on the language model’s memory. Consistent with this principle, the proposed framework maintains awareness of previously discussed evidence, dialogue history, and investigation progress to preserve narrative continuity throughout the interaction.

3.2.3 Dialogue Policies and Narrative Progression

Dialogue policies regulate how conversations progress by determining appropriate responses based on the current dialogue state. Instead of allowing unrestricted response generation, dialogue policies guide conversational flow toward predefined objectives while maintaining contextual coherence. Recent research demonstrates that dialogue policies combined with dialogue state tracking effectively regulate large language models without requiring additional model training (Zhang et al., 2023). These findings support the Narrative Progression component of the proposed Dialogue Flow Manager, which controls the pace of Sherlock Holmes’ investigation, maintains the logical sequencing of evidence, and reduces persona drift by applying dialogue constraints before response generation.

3.3 Interaction Steering Theory

Interaction steering refers to the use of external mechanisms to regulate how a conversational agent processes user interactions and generates responses throughout a dialogue. Rather than relying solely on the reasoning capabilities of a language model, interaction steering employs structured prompts, dialogue constraints, and validation mechanisms to improve consistency and maintain conversational objectives without modifying the model itself.

Recent studies show that instruction-tuned large language models can be effectively guided through external orchestration techniques, particularly in role-playing applications where maintaining persona consistency over extended conversations is challenging (Z. M. Wang et al., 2024; Zhou et al., 2024). Guided by these principles, the proposed Interaction-Layer Middleware Framework regulates interactions before response generation and validates responses afterward through structured prompting, input constraints, dialogue management, and response validation.

3.3.1 Prompt Engineering

Prompt engineering is the practice of designing structured prompts that guide a language model toward producing responses aligned with a specific objective. By providing contextual information, role descriptions, and explicit instructions, prompt engineering enables language models to perform specialized tasks without additional training.

Research has shown that structured prompting improves response quality and consistency. The CLARIFIES+ Framework demonstrates that prompts with clearly defined roles, objectives, and contextual information produce more reliable outputs than unstructured prompts (Barrot, 2026). Similarly, role-playing studies report that explicitly defining character identity improves persona fidelity (Z. M. Wang et al., 2024). Despite these benefits, prompt engineering alone is insufficient for maintaining persona consistency during long conversations because the influence of the initial prompt gradually weakens, increasing the likelihood of persona drift (Samuel et al., 2025; Zhou et al., 2024). Consequently, the proposed framework employs a Prompt Builder that continuously updates the structured prompt using conversation history, dialogue state, investigative context, and dialogue constraints.

3.3.2 Constraint-Based Interaction Steering

Constraint-based interaction steering extends prompt engineering by applying pre-defined behavioral constraints that regulate conversation flow. Instead of relying solely on an initial system prompt, the language model is guided through dialogue constraints that preserve contextual consistency, narrative progression, and persona fidelity.

Research shows that combining dialogue management with interaction constraints produces more coherent multi-turn conversations than static prompting alone. Frameworks such as SGP-TOD and dialogue state tracking approaches demonstrate that maintaining structured dialogue states improves conversational coherence and reduces context loss (Khan et al., 2025; Zhang et al., 2023). Likewise, role-playing studies consistently report that external interaction management mechanisms are more effective than prompt engineering alone in maintaining persona consistency (Samuel et al., 2025; Z. M. Wang et al., 2024; Zhou et al., 2024).

Based on these principles, the proposed framework implements a Dialogue Constraint Library, where dialogue constraints are selected according to the cur-

rent dialogue state and the Judge LLM’s evaluation before being injected into the final prompt. This enables the system to maintain Sherlock Holmes’ persona, preserve narrative continuity, and regulate conversation progression while allowing the language model to generate natural responses.

3.4 Middleware Architecture

Middleware is a software layer positioned between distributed system infrastructure and application software. Bernstein (Bernstein, 1996) describes middleware as distributed system services that provide programming interfaces and protocols to support communication between applications and distributed environments. Similarly, Campbell et al. (Campbell, Coulson, & Kounavis, 1999) define middleware as a software layer above distributed system infrastructure and below the application layer. Middleware is used to manage the complexity of distributed systems by providing common services that simplify application development (Bernstein, 1996; Campbell et al., 1999).

The middleware approach allows applications to interact with underlying system resources without directly handling all implementation details of the distributed environment. By providing an intermediate layer between applications and infrastructure, middleware separates application functionality from system-level concerns and improves the management of complex software environments (Bernstein, 1996; Campbell et al., 1999).

In recent years, the middleware concept has also been applied to large language model (LLM) systems. LLM applications often involve additional components beyond the language model itself, such as external services, retrieval systems, and supporting software components. As a result, recent studies have explored middleware architectures that coordinate interactions between the language model and surrounding system components (Gu et al., 2024; Guran, Knauf, Ngo, Petrescu, & Rellermeyer, 2024).

The proposed framework adopts this middleware principle by introducing an Interaction Layer that operates as an intermediate layer around the response generation process. Similar to existing LLM middleware approaches, the Interaction Layer separates additional interaction management responsibilities from the language model, allowing the model to focus on generating responses.

3.4.1 Middleware in Large Language Model Systems

Middleware in LLM systems extends the traditional idea of an intermediate software layer by coordinating processes surrounding language model execution. Instead of requiring the language model to directly manage all interactions with external components, middleware can provide additional processing and coordination between the model and other parts of the system.

Gu et al. (2024) introduce middleware as a layer that assists language models in interacting with complex environments. Their work presents middleware tools that support interactions with external resources, such as knowledge bases and databases, allowing the language model to operate without directly managing the complexity of these environments (Gu et al., 2024).

Guran et al. (2024) propose a middleware architecture for LLM-based applications that addresses challenges related to deploying and integrating LLM services into existing software systems. Their work describes middleware as an architectural layer that manages the components and interactions required for LLM deployment (Guran et al., 2024).

These studies show that middleware can be used to coordinate processes around an LLM while keeping response generation as the responsibility of the language model. This architectural principle motivates the use of an Interaction Layer in the proposed framework.

3.4.2 Layered Software Architecture

Layered software architecture organizes a software system into a hierarchy of layers with assigned responsibilities and controlled interactions. Marmsoler, Malkis, and Eckhardt (2015) describe layered architecture as a hierarchy in which each layer provides services to the layer above while using services from the layer below through defined interfaces. Similarly, Savolainen and Myllärniemi (2009) explain that layered architectures separate software into distinct layers, each responsible for a specific set of functions within the system.

By separating responsibilities across architectural layers, this design supports separation of concerns, reduces coupling between software components, and limits direct dependencies between different parts of the system (Marmsoler et al., 2015; Savolainen & Myllärniemi, 2009). Because each layer performs a specific function, changes within one layer can be made without requiring changes to other layers, provided that the interfaces between layers remain consistent.

This architectural principle provides the basis for organizing the proposed framework into separate layers responsible for interaction management, response generation, and response validation. Rather than assigning all conversational responsibilities to a single language model, the layered architecture separates these major functions into distinct architectural layers that communicate through well-defined interfaces. This separation of concerns reduces dependencies between interaction management and language model execution, allowing conversation flow management and response validation to be performed independently of the language model, while the language model focuses solely on generating responses.

3.4.3 Modular Software Design

Modular software design organizes a software system into separate modules, where each module is responsible for a specific aspect of the system. Parnas (1972) proposed decomposing software according to design decisions that are likely to change independently, allowing each module to conceal its internal implementation while exposing interfaces to the rest of the system.

This approach allows individual modules to be modified, replaced, enabled, or disabled without requiring changes throughout the system, provided that the interfaces between modules remain consistent (Parnas, 1972). Separating software into modules also allows individual components to be tested independently before they are integrated into the complete system.

This principle provides the basis for organizing the proposed framework into multiple software modules. By assigning different responsibilities to separate modules, the framework separates persona construction, interaction management, response generation, response validation, and interaction logging into independent software modules that communicate through defined interfaces. This organization also supports systematic experimentation by allowing individual middleware modules to be enabled or disabled while preserving the remaining architecture. As a result, the contribution of different middleware mechanisms can be evaluated independently under consistent experimental conditions.

3.5 Response Validation Theory

While prompt engineering and constraint-based interaction steering regulate a language model’s behavior before a response is produced, they cannot guarantee that every generated output actually satisfies the intended persona, safety, and

narrative-continuity requirements. Response validation theory addresses this gap by treating each model output as a candidate response that must be verified after generation and before it reaches the user. Rather than relying solely on upstream instructions to shape behavior, a validation layer inspects the realized response against explicit reference criteria, allowing the system to detect persona drift, contradictions, or policy violations that upstream constraints failed to prevent.

This reactive checkpoint complements the proactive mechanisms established in Sections 3.3 and 3.4, forming the second half of a closed-loop interaction cycle: steer before generation and verify after generation.

3.5.1 LLM-as-a-Judge Evaluation

A central approach to automated post-generation evaluation is the use of a separate language model as a judge function that scores a candidate response against a reference, such as the original persona prompt or prior dialogue turns. Abdulhai et al. (2025) formalize this through three complementary consistency metrics: prompt-to-line consistency, which checks whether an utterance aligns with the initial persona description; line-to-line consistency, which detects contradictions between an utterance and earlier turns in the same dialogue; and Q&A consistency, which probes whether the underlying beliefs and traits implied by a response remain stable when queried indirectly.

Critically, their validation study found that an LLM judge achieved higher agreement with human annotators than human annotators achieved with one another, supporting the use of LLM-as-a-Judge as a reliable and scalable substitute for exhaustive manual review. A related evaluation strategy is reported by Z. M. Wang et al. (2024), who pair an LLM evaluator with ROUGE-L overlap scores and human ratings to assess role-playing outputs along speaking-style imitation, response accuracy, and role-specific knowledge, using pairwise win-rate comparisons rather than absolute scoring.

Together, these studies indicate that a judge model can assess persona fidelity along multiple, separable dimensions rather than treating consistency as a single binary property.

3.5.2 Guardrail-Based Output Filtering

A second, complementary strand of response validation treats generated output as content to be filtered for policy compliance rather than persona fidelity. Shen,

Chen, Backes, Shen, and Zhang (2024) evaluate three such mechanisms operating downstream of an LLM’s built-in alignment: a moderation endpoint that classifies responses into predefined harm categories, a fine-tuned moderation classifier that labels question–response pairs by risk level, and a guardrail toolkit that screens inputs and outputs against usage policies before a response is delivered.

Their findings show that classification-based guardrails can measurably reduce the success rate of adversarial or policy-violating prompts without modifying the underlying model, but that this protection is partial and can be circumvented through paraphrasing. This indicates that filtering mechanisms alone are an incomplete substitute for consistency-oriented judging. This distinction motivates treating safety filtering and persona-consistency verification as separate, non-substitutable stages of response validation, since a response may be policy-compliant while still breaking character, or persona-consistent while still containing unsafe content.

3.5.3 Response Verification in the Proposed Framework

Drawing on these two validation paradigms, the proposed Output Validation Module operationalizes response verification as a two-stage check performed on every candidate response before delivery. The Judge LLM introduced in Section 3.5.1 first evaluates the candidate against the persona baseline established by the five-layer Persona Object Model in Section 3.1 and the current dialogue state maintained by the Dialogue State Tracking component in Section 3.2.1, following the consistency-scoring approach of Abdulhai et al. (2025).

Responses that fail this check are flagged for regeneration or constraint re-injection through the Dialogue Constraint Library rather than being passed directly to the user. A secondary filtering pass, informed by the guardrail evaluations of Shen et al. (2024), screens the response for content that falls outside the fictional persona’s established knowledge and behavioral boundaries.

By separating persona-consistency verification from safety filtering, the framework ensures that response validation addresses both dimensions of failure identified in this section, closing the loop between interaction steering and response generation within the middleware architecture described in Section 3.4.

Chapter 4

Persona Agent Construction and Validation

Figure 4.1 shows the architectural overview of the proposed framework. The system utilizes three primary agents: the Persona Construction Agent, the Input Constraint Engine, and the Output Validation Engine. The Persona Construction Agent handles the parsing of canonical literary sources, the extraction of behavioral traits, and the independent examination of the finalized persona profile. The Input Constraint Engine and the Output Validation Engine serve as the active middleware, responsible for evaluating user inputs, structuring prompts, and verifying generated responses to prevent persona drift during live multi-turn interactions. Because the interaction middleware relies entirely on the accuracy of the baseline character data, this chapter presents the Persona Construction Agent in detail, outlining its data source, extraction methodology, and internal validation procedure.

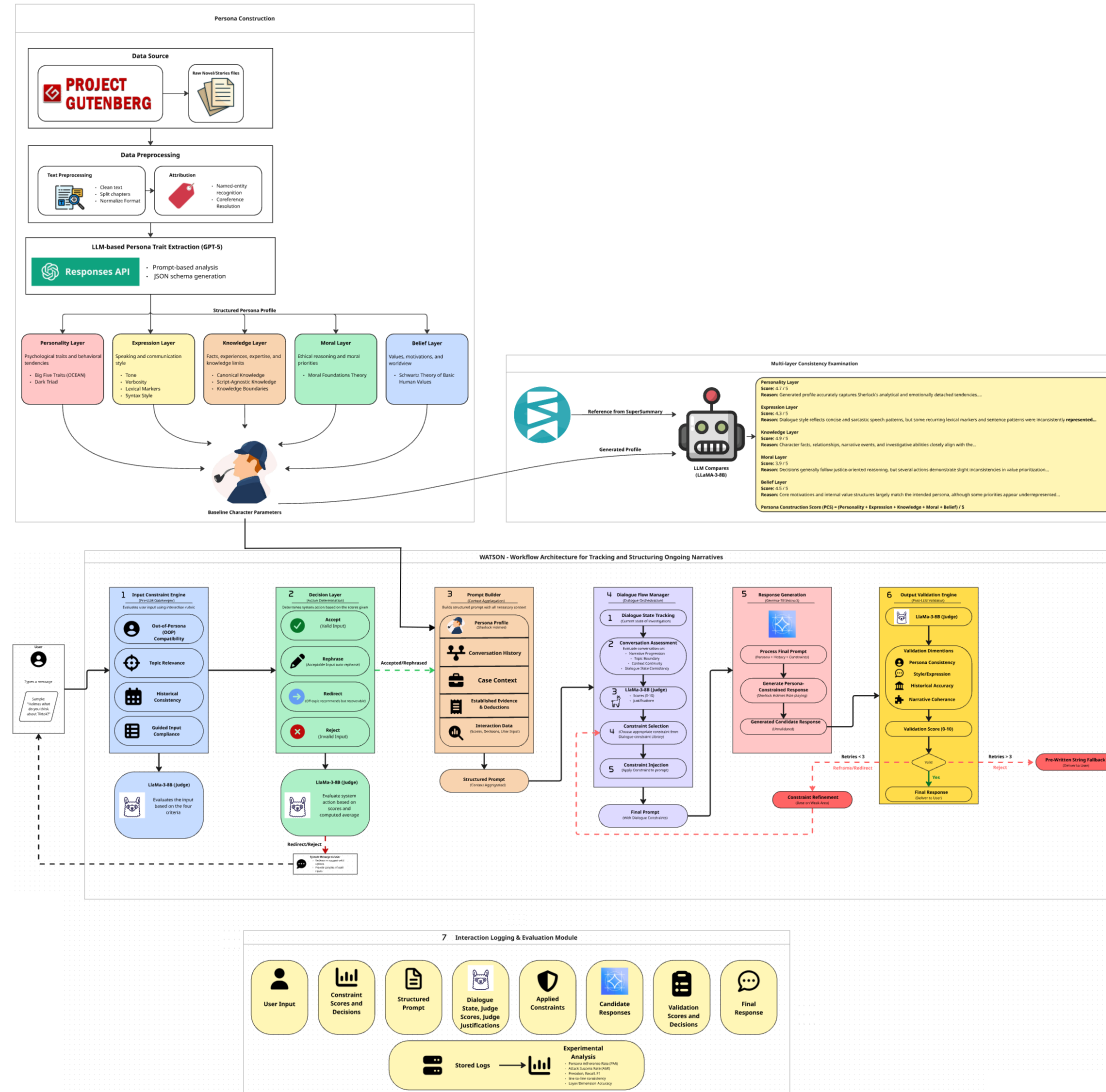


Figure 4.1: Proposed System Framework

The Persona Construction Methodology defines the sequence of processes used to construct the baseline persona profile of Sherlock Holmes prior to system deployment. The methodology consists of three sequential stages: data preparation, persona trait extraction, and persona validation. These stages collectively transform the canonical literary corpus into a validated, structured persona profile that serves as the behavioral foundation for the succeeding software modules within the proposed framework. Figure 4.1 illustrates the complete persona construction pipeline implemented by the proposed framework.

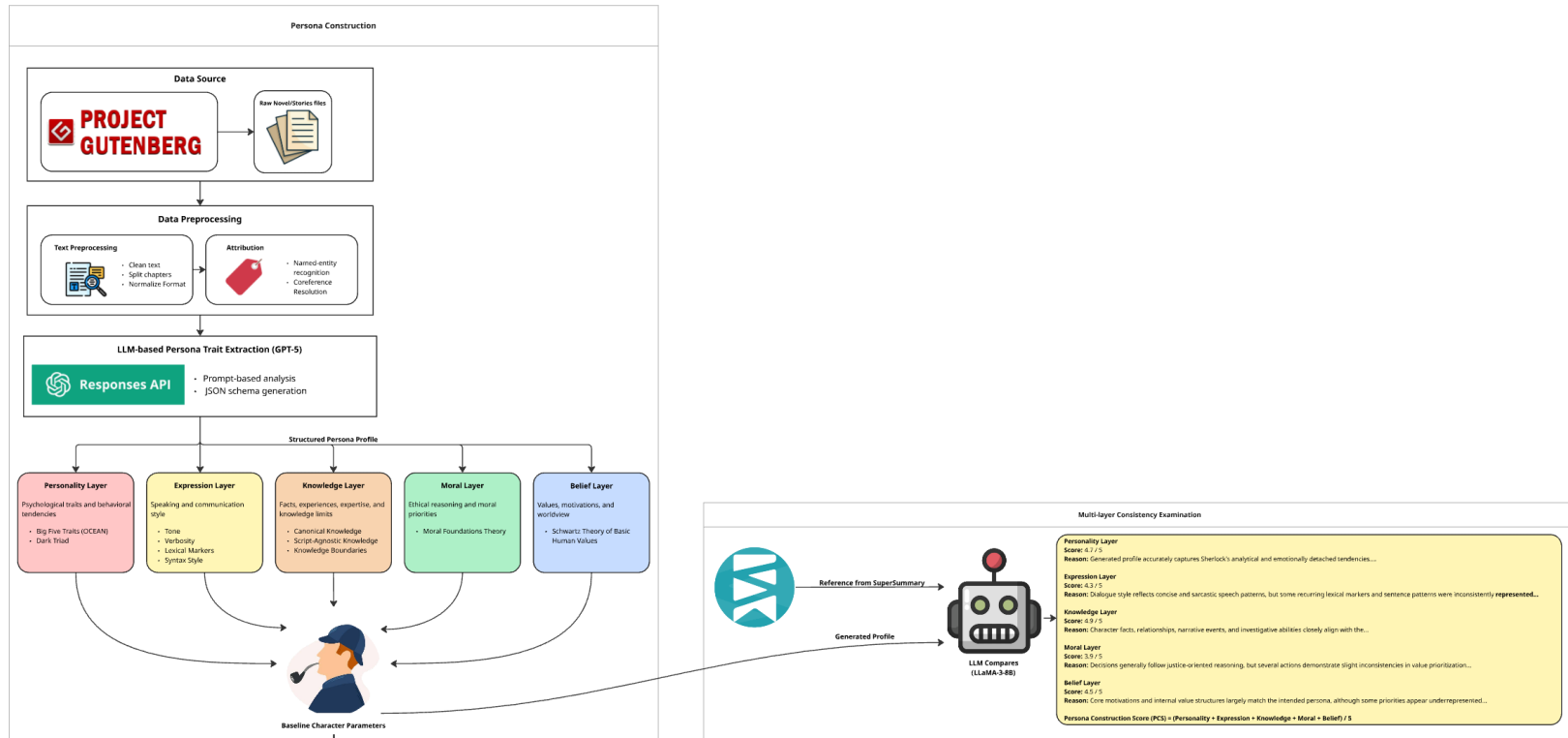


Figure 4.2: Persona Construction Component

4.1 Data Preparation

The initial phase of the persona construction pipeline focuses on the acquisition and refinement of the foundational text. The succeeding subsections detail the specific canonical source material collected for the study, followed by the natural language processing techniques applied to transform the raw literary corpus into a structured format optimized for behavioral extraction.

4.1.1 Data Source

The first stage of persona construction involves assembling the textual corpus that serves as the factual grounding for Sherlock Holmes’ character. Because the objective of the study is to emulate the canonical literary portrayal, only the original Sherlock Holmes books written by Sir Arthur Conan Doyle are included.

The collected corpus consists of canonical Sherlock Holmes literary works obtained from the Sir Arthur Conan Doyle author collection available through Project Gutenberg (Project Gutenberg, n.d.-a). According to the Project Gutenberg License, unrestricted works may be used “for nearly any purpose such as creation of derivative works, reports, performances and research” (Project Gutenberg, n.d.-b). These literary works are therefore used in accordance with their public-domain status and the applicable terms of use solely for academic, non-commercial research purposes involving computational analysis and persona extraction. The corpus contains the raw narrative data, including Holmes’ dialogue, investigative methodologies, interpersonal interactions, and behavioral patterns, that collectively define his fictional identity. Utilizing strictly canonical sources minimizes the inconsistencies that frequently occur when secondary adaptations or modern reinterpretations are incorporated into the dataset. Consequently, the extracted persona remains tightly aligned with Doyle’s original characterization (Z. M. Wang et al., 2024; Zhou et al., 2024). Once compiled, the corpus is forwarded to the data preparation stage for preprocessing.

4.1.2 Data Preprocessing

Before persona information can be computationally extracted, the literary corpus undergoes a preprocessing stage to standardize formatting and facilitate automated analysis. During this stage, the textual data is cleaned by removing extraneous metadata, correcting formatting inconsistencies, and explicitly separating narrative passages from dialogue. Natural language processing (NLP) techniques,

specifically Named Entity Recognition (NER) and coreference resolution, are applied to systematically identify entity occurrences and track speaker attribution across the text.

Coreference resolution is particularly necessary in literary narratives, where characters are frequently referenced using third-person pronouns (e.g., "he," "his") or descriptive phrases (e.g., "the detective," "my friend") rather than their explicit proper names. If unparsed text is passed directly to a downstream large language model (LLM), pronouns like "he" can create ambiguity as to whether the passage refers to Sherlock Holmes or another character, such as Dr. Watson. Explicitly resolving these coreference chains beforehand ensures that character attribution is non-ambiguous when processed by the LLM. This design choice aligns with empirical findings by (Amalvy, Labatut, & Dufour, 2025), who demonstrated that relying solely on NER misses a significant portion of character mentions, and that coreference resolution is required to capture these occurrences across a literary text.

The primary objective of this preprocessing stage is to convert the unstructured corpus into a structured dataset suited for language model analysis. Resolving entity references up front reduces downstream extraction errors and improves the reliability of the behavioral information extracted in subsequent steps (Z. M. Wang et al., 2024; X. Wang et al., 2024).

4.2 Persona Trait Extraction

Following data preparation, the framework extracts the core behavioral matrix from the processed corpus. This is achieved using GPT-5 through the OpenAI Responses API, which analyzes the processed literary documents to identify characteristics that consistently define Sherlock Holmes. The OpenAI API is used in accordance with its applicable terms of use and solely for academic, non-commercial research purposes. The prepared texts are supplied to the language model together with highly specific extraction prompts that instruct it to identify behavioral evidence corresponding to each behavioral dimension of the proposed persona representation. Rather than generating conversational responses, the model functions exclusively as an information extraction tool. This approach follows the structured persona extraction methodology adopted by Park et al. (2025), who employed GPT-4 Turbo through the OpenAI Assistants API to extract and organize character traits into structured persona representations. However, since the Assistants API has been superseded by the OpenAI Responses API, the present study utilizes GPT-5 while retaining the same structured extrac-

tion methodology.

To ensure that the extracted data is programmatically reliable, the framework leverages the API’s Structured Outputs capability to enforce strict compliance with a predefined JSON schema. This converts the qualitative behavioral information into a structured, machine-readable object. The extracted traits are mapped directly into the framework’s five behavioral dimensions:

- Personality
- Expression
- Knowledge
- Moral
- Belief

Automating this extraction improves architectural consistency by deriving traits directly from verifiable literary evidence rather than relying on manually authored, subjective prompts. Furthermore, categorizing the data into these predefined JSON dimensions simplifies its eventual integration into the middleware layer. The resulting unvalidated profile is then forwarded directly to the final stage of the pipeline: the independent persona validation procedure, which is executed and evaluated as detailed in Section 4.3.

4.2.1 Persona Trait Extraction Component

The Persona Trait Extraction Component serves as the primary data processing engine responsible for converting unstructured literary text into a structured, machine-readable format (Zhu et al., 2024). Within the system architecture, this component interfaces directly with the OpenAI API, utilizing its Structured Outputs capabilities to enforce strict JSON schema compliance during generation.

By supplying the preprocessed literary corpus alongside specific extraction prompts, this component tasks the underlying language model with parsing narrative and dialogue passages to identify core behavioral markers, an approach proven effective for automated character extraction from canonical sources (Z. M. Wang et al., 2024; Zhou et al., 2024). The component does not generate new conversational text; rather, it performs targeted information extraction, categorizing the identified traits into the five predefined behavioral dimensions required by the

Multi-dimensional Persona Representation. The output of this component is a highly structured JSON object that cleanly disentangles the character’s identity, serving as the unvalidated baseline profile.

4.2.2 Multi-dimensional Persona Representation

The proposed framework represents Sherlock Holmes’ identity using a multi-dimensional persona representation composed of five behavioral layers. Instead of storing persona information as a single block of text, the proposed framework separates different behavioral characteristics into independent dimensions. This modular representation enables subsequent software components to reference specific aspects of the character according to their respective responsibilities.

The five behavioral dimensions are:

- Personality Layer
- Expression Layer
- Knowledge Layer
- Moral Layer
- Belief Layer

The succeeding subsections describe each layer in greater detail.

Personality Layer

The Personality Layer defines Sherlock Holmes’ characteristic patterns of thinking and behavior based on the psychometric frameworks established in Section 3.1.1. It describes stable behavioral traits that remain consistent regardless of the specific investigation. Examples of information represented within this layer include Holmes’ analytical reasoning, emotional restraint, and high conscientiousness, representing the Five-Factor Model, alongside necessary adversarial traits such as arrogance or manipulative tendencies when dealing with suspects, representing the Dark Triad. This layer serves as the primary behavioral reference used to preserve Holmes’ overall personality throughout the conversation (Z. M. Wang et al., 2024; Zhou et al., 2024).

Expression Layer

The Expression Layer describes how Sherlock Holmes communicates with other characters. Utilizing the stylometric parameters explicitly defined in Section 3.1.2, this layer captures specific linguistic markers rather than behavioral traits (N. Wang et al., 2024). These parameters include his highly formal vocabulary selection, deductive sentence structure, authoritative conversational tone, and proper Victorian syntax. Maintaining these precise lexical markers allows the generated responses to accurately reflect Holmes’ distinctive speaking style while preventing the generation model from regressing to a default AI tone.

Knowledge Layer

The Knowledge Layer defines the factual and epistemological boundaries of Sherlock Holmes’ expertise. Mapped according to the structural constraints detailed in Section 3.1.3, this layer explicitly contains Holmes’ canonical investigative methods, his Victorian-era scientific knowledge, and his relational topologies regarding other characters (Zhu et al., 2024). By explicitly defining these specific domain boundaries, the framework reduces the likelihood of anachronistic hallucinations during role-playing conversations, adhering to the principles of role-specific memory isolation (Park et al., 2025; N. Wang et al., 2024).

Moral Layer

The Moral Layer represents Sherlock Holmes’ ethical principles and decision-making standards. Following the Moral Foundations Theory (MFT) baseline anchored in Section 3.1.4, this layer codifies his specific moral axes (Abdulhai et al., 2023). Examples include his commitment to justice over strict legal procedure, his absolute respect for factual evidence, and his willingness to pursue the truth regardless of personal difficulty. Structuring these specific principles contributes to consistent behavioral responses during morally complex conversational scenarios.

Belief Layer

The Belief Layer represents Sherlock Holmes’ underlying worldview and long-term convictions. Structured around the fundamental value dimensions outlined in Section 3.1.5, this layer details Holmes’ intrinsic motivations (Yao et al., 2023). Unlike

the Moral Layer, which focuses on ethical decision-making, this layer describes his high valuation of self-direction, his strict reliance on empirical observation over emotion, and his relentless pursuit of intellectual stimulation. Together with the remaining behavioral dimensions, this layer completes the structured persona representation used throughout the framework.

4.3 Persona Validation Procedure

Following persona trait extraction, the generated persona profile undergoes an independent validation procedure before it is incorporated into the proposed framework. The purpose of this stage is to verify that the extracted behavioral representation accurately reflects Sherlock Holmes’ characterization and is suitable for use by the succeeding software modules. Consistent with the character profiling evaluation framework proposed by Yuan, Chen, Zhao, et al. (2024), the proposed framework adopts an LLM-as-a-Judge approach in which an independent language model evaluates the factual consistency of the generated persona profile against a reference profile.

The validation procedure consists of three stages. First, the Persona Trait Extraction Component generates a structured JSON persona profile from the pre-processed literary corpus. Second, a reference profile of Sherlock Holmes is obtained from SuperSummary, following the reference profile methodology of Yuan et al. (2024), which uses expert-authored SuperSummary character analyses as the reference for evaluating generated character profiles. Finally, an independent Judge LLM compares the generated persona profile with the reference profile and evaluates each behavioral dimension before the profile is accepted for use within the proposed framework.

Figure 4.3 illustrates the persona validation procedure. The generated persona profile and the SuperSummary reference profile are provided to the Judge LLM, which evaluates their factual consistency across the five behavioral dimensions. The Judge LLM then assigns Consistency Scores and provides qualitative feedback, which are used to determine whether the generated persona profile satisfies the acceptance criteria before being incorporated into the proposed framework.

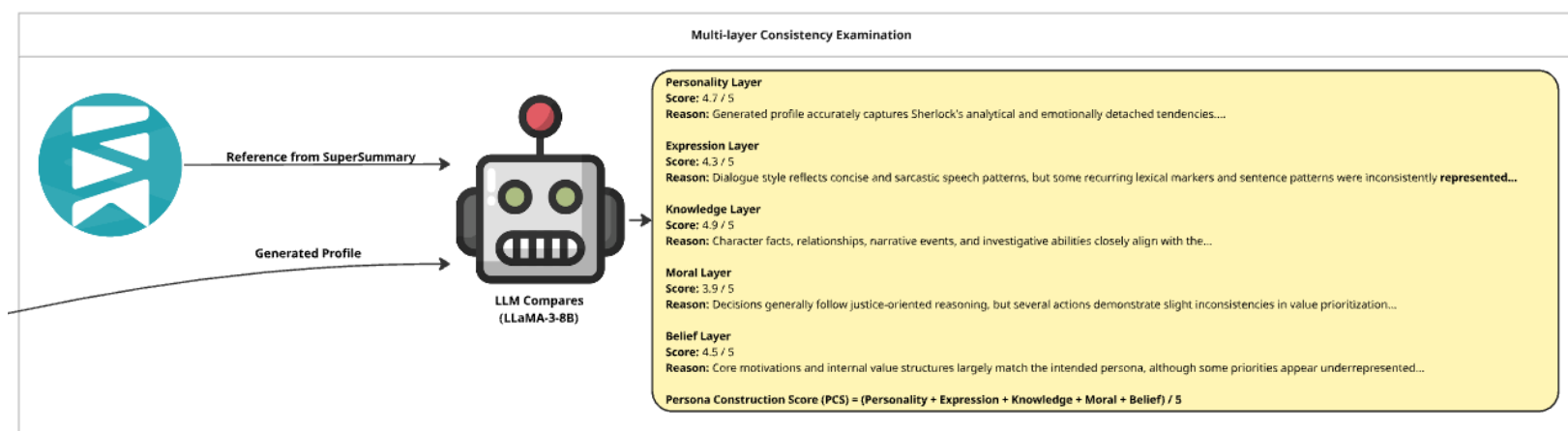


Figure 4.3: Persona Validation Component

4.3.1 LLM Evaluation

An independent Judge LLM, specifically LLaMA-3-8B Instruct, is employed solely to evaluate the generated persona profile and does not participate in the persona extraction process. The Judge LLM receives the generated **JSON** persona profile together with the reference profile and evaluates the factual consistency of the extracted behavioral information.

Consistent with the evaluation protocol of Yuan et al. (2024), the Judge LLM independently evaluates each of the five behavioral dimensions and assigns a Consistency Score ranging from 1 to 5, where higher scores indicate greater factual consistency between the generated and reference profiles. While Yuan et al. (2024) evaluate the dimensions of attributes, relationships, events, and personality, the proposed framework applies the same evaluation methodology to the five behavioral dimensions defined in Section 4.2.2.

In addition to assigning a numerical score, the Judge LLM provides qualitative feedback describing any missing information, factual inaccuracies, or inconsistencies identified within each behavioral dimension. These evaluations are used to determine whether the generated persona profile accurately represents Sherlock Holmes before it is integrated into the middleware framework.

4.3.2 Validation Criteria

Each behavioral dimension is evaluated using a five-point Consistency Score. Following Yuan et al. (2024), the Judge LLM assigns a score ranging from 1 (lowest) to 5 (highest), where higher scores indicate greater factual consistency between the generated persona profile and the reference profile. The proposed framework applies the same scoring scale to evaluate the five behavioral dimensions of Sherlock Holmes: Personality, Expression, Knowledge, Moral, and Belief.

An overall **Persona Construction Score (PCS)** is computed by averaging the Consistency Scores assigned to the five behavioral dimensions:

$$PCS = \frac{Personality + Expression + Knowledge + Moral + Belief}{5} \quad (4.1)$$

Although Yuan et al. (2024) do not establish a formal acceptance threshold, their highest-performing character profiling methods consistently achieved average Consistency Scores of approximately 3.6 to 4.0. Based on these reported results,

the present study adopts a minimum Persona Construction Score of 3.5 as the acceptance criterion. This specific threshold aligns with precedent from prior research that uses a similar validation procedure, while pragmatically considering the differences in evaluation setup and available resources between those baseline studies and the current framework. Persona profiles achieving a PCS of 3.5 or higher are considered to exhibit satisfactory factual consistency with the reference profile and are accepted for integration into the proposed middleware framework.

If a persona profile fails to meet the acceptance threshold, the qualitative feedback generated by the Judge LLM is leveraged to diagnose specific areas requiring improvement. Refinement strategies may include tuning the prompt instructions, adjusting extraction parameters, or enhancing document preprocessing pipelines. The persona profile is then iteratively regenerated and re-evaluated until it achieves the required consistency score. Once validated, the final structured JSON persona profile serves as the baseline representation of Sherlock Holmes for integration into the middleware framework.

4.3.3 Final Persona Profile

Upon completion of the validation procedure, the generated persona profile is reviewed using the evaluation results produced by the Judge LLM. Profiles that achieve the required Persona Construction Score (PCS) are accepted as the baseline representation of Sherlock Holmes within the proposed framework. This validated persona profile serves as the primary knowledge source for all succeeding software modules, including interaction management, prompt construction, response generation, and output validation.

Conversely, if the generated persona profile exhibits substantial inconsistencies or omissions, the persona construction process is repeated. Refinements may include modifying the extraction prompts, adjusting the preprocessing pipeline, or incorporating additional canonical textual evidence before repeating the validation procedure. This iterative process continues until the generated persona profile achieves the required Persona Construction Score (PCS).

The final validated persona profile is stored as a structured JSON object organized according to the five behavioral dimensions defined in Section 4.2.2. Because this profile remains unchanged throughout system deployment, it provides a stable behavioral baseline against which conversational responses can subsequently be generated and evaluated, thereby supporting the objective of maintaining Sherlock Holmes’ persona consistency during extended multi-turn interactions.

Chapter 5

WATSON - Workflow Architecture for Tracking and Structuring Ongoing Narratives

The previous chapter presented the design and construction of the Persona Knowledge Base, which captures Sherlock Holmes' personality, communication style, knowledge, moral principles, and beliefs from Arthur Conan Doyle's original works. The resulting structured persona profile serves as the foundational knowledge of the chatbot and, as illustrated in the overall system architecture in Figure 4.1, is passed to the proposed WATSON (Workflow Architecture for Tracking and Structuring Ongoing Narratives) Interaction-Layer Middleware. While the Persona Knowledge Base defines the character's identity, the middleware is responsible for applying this information throughout the conversation by regulating user interactions, managing dialogue flow, and validating generated responses to maintain persona consistency during extended role-playing interactions. This chapter discusses the architecture of the proposed middleware and the evaluation mechanisms that enable these interaction management processes.

5.1 Interaction Layer Architecture

The proposed Interaction-Layer Middleware (WATSON) serves as the central processing framework that manages every interaction between the user and the response generation model. While the Persona Knowledge Base developed in Chapter 4 defines Sherlock Holmes' personality, knowledge, beliefs, and communication style, the middleware is responsible for applying this information throughout the

conversation. Instead of forwarding user messages directly to the language model, the middleware regulates the interaction before and after response generation to preserve persona consistency during extended conversations (Budagam, 2024; Zhang et al., 2023).

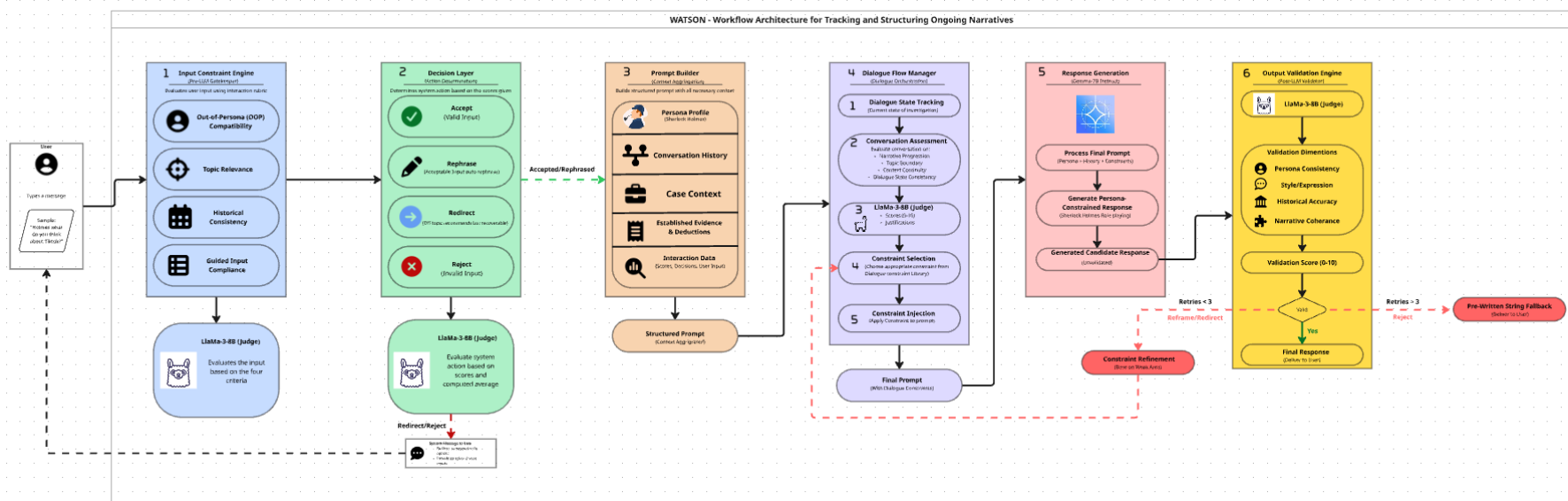


Figure 5.1: WATSON Interaction-Layer Middleware Architecture

The middleware consists of six interconnected components that operate sequentially throughout every conversational turn: the Input Constraint Engine, Decision Layer, Prompt Builder, Dialogue Flow Manager, Response Generation Layer, and Output Validation Engine. Each component performs a specific responsibility within the interaction pipeline, allowing user interactions to be evaluated, contextualized, processed, and verified before a response is presented to the user.

Unlike conventional chatbot architectures that rely primarily on the language model to manage both conversation flow and response generation, the proposed framework separates these responsibilities across independent middleware components. This modular design enables interaction management, dialogue control, and response validation to operate independently of the response generation model, providing a more structured approach to maintaining persona consistency throughout multi-turn conversations (Samuel et al., 2025; X. Wang et al., 2024).

5.1.1 Input Constraint Engine

The Input Constraint Engine is the first processing component of the proposed Interaction-Layer Middleware. Its primary responsibility is to evaluate every user interaction before it reaches the response generation model. Rather than allowing all user messages to be processed directly by the language model, the middleware first determines whether the interaction is appropriate for the Sherlock Holmes role-playing environment. This approach allows the framework to identify interactions that may disrupt persona consistency or the intended investigative narrative before response generation begins (Das & Fioretto, 2026; Kumar et al., 2026).

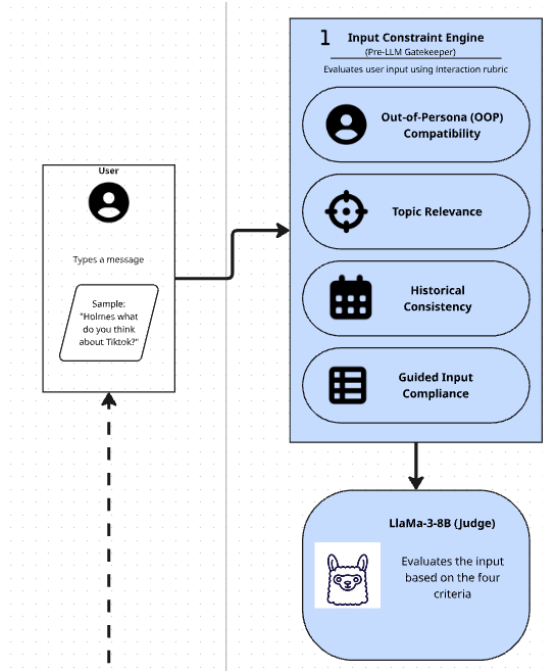


Figure 5.2: Input Constraint Engine Architecture

As shown in Figure 5.2, the Input Constraint Engine receives the user’s message together with the current conversation context. These inputs are forwarded to LLaMA-3-8B Instruct, which functions as the Judge LLM for interaction evaluation. Instead of generating a conversational response, the Judge LLM analyzes the interaction and determines whether it is compatible with the objectives of the role-playing system.

The evaluation considers four aspects of the interaction: the user’s compatibility with Sherlock Holmes’ persona, the relevance of the topic to the ongoing conversation, the historical consistency of the request, and whether the interaction follows the intended use of the chatbot. Rather than making the final decision itself, the Input Constraint Engine produces a structured evaluation containing individual criterion scores and supporting justifications. These evaluation results are forwarded to the Decision Layer for further processing.

5.1.2 Decision Layer

The Decision Layer is responsible for interpreting the evaluation results produced by the Input Constraint Engine and determining how the interaction should proceed. Rather than treating every interaction as either valid or invalid, the pro-

posed framework supports multiple decision outcomes that allow the middleware to respond more appropriately to different types of user inputs.

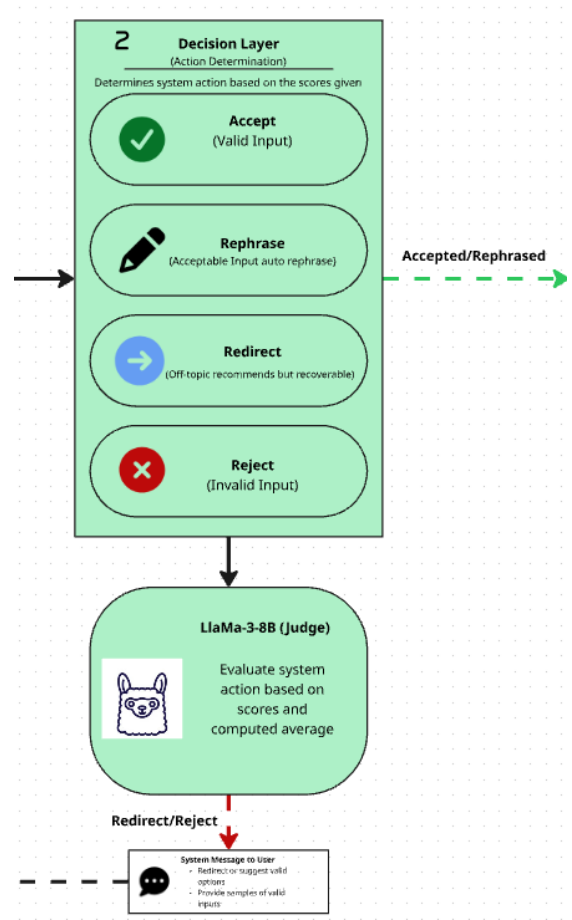


Figure 5.3: Decision Layer Architecture

The Decision Layer receives the structured evaluation generated by the Input Constraint Engine and compares the results against predefined decision criteria. Based on this assessment, the middleware determines whether the interaction should continue without modification, be automatically rephrased, be redirected toward a more appropriate topic, or be rejected. This additional processing stage provides greater flexibility than a traditional binary filtering approach by allowing recoverable interactions to remain part of the conversation.

If the interaction satisfies the required conditions, it proceeds directly to the Prompt Builder. Interactions that require clarification or refinement are automatically rephrased before continuing through the pipeline. Requests that fall outside the intended scope of the role-playing environment may be redirected to-

ward a more suitable conversation, while interactions that cannot reasonably be incorporated into the system are rejected before reaching the response generation model.

5.1.3 Prompt Builder

The Prompt Builder is responsible for constructing the structured prompt submitted to the response generation model. Rather than forwarding only the user’s message, the Prompt Builder combines multiple sources of information to provide the language model with sufficient context for generating a persona-consistent response.

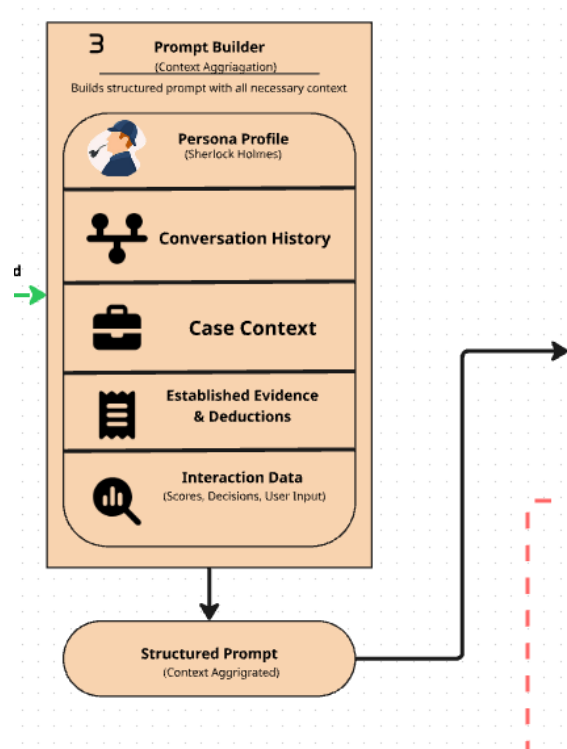


Figure 5.4: Prompt Builder Architecture

The Prompt Builder receives the validated user input from the Decision Layer together with the Persona Knowledge Base developed in Chapter 4. It also retrieves the current conversation history, dialogue state, and investigation context maintained by the middleware. These components are combined into a structured prompt that serves as the complete input for the response generation model.

5.1.4 Dialogue Flow Manager

The Dialogue Flow Manager is responsible for maintaining the progression of the conversation after the structured prompt has been constructed. While the Prompt Builder prepares the information required by the language model, the Dialogue Flow Manager ensures that the conversation develops in a logical and consistent manner throughout multiple dialogue turns. This approach follows dialogue management techniques that regulate conversational flow using explicit dialogue states rather than relying solely on the language model’s internal context (Zhang et al., 2023).

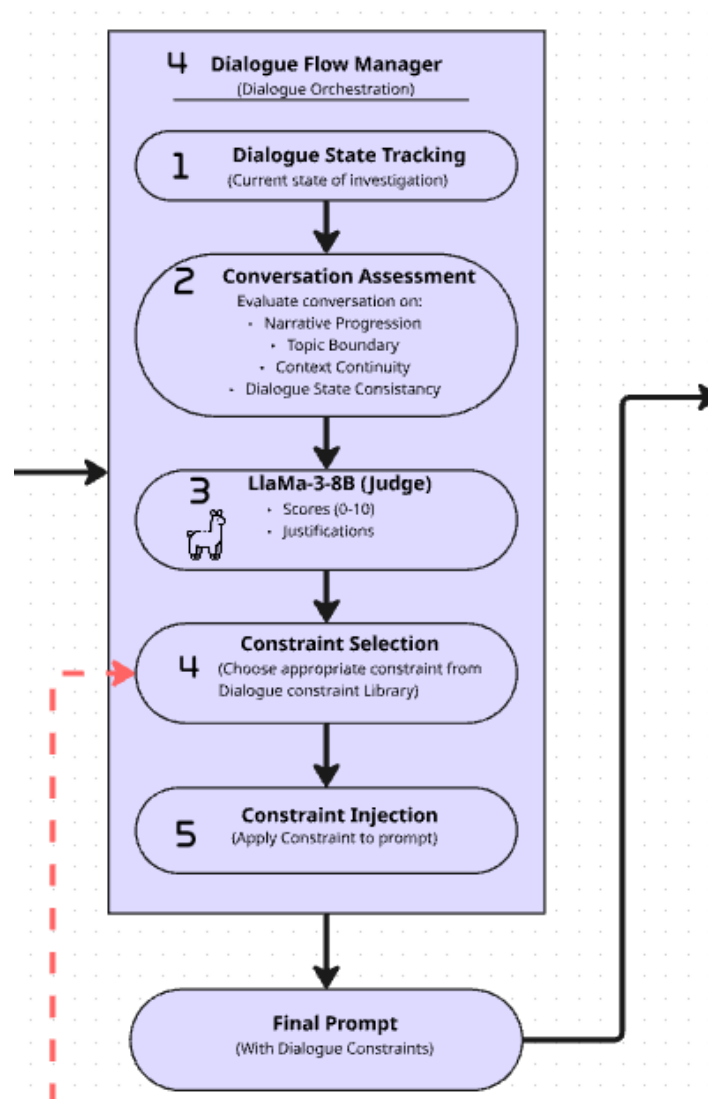


Figure 5.5: Dialogue Flow Manager Architecture

The Dialogue Flow Manager receives the structured prompt from the Prompt Builder together with the current conversation state maintained by the middleware. Using this information, the component determines how the investigation should progress while ensuring that the generated response remains consistent with Sherlock Holmes’ reasoning process and the established narrative.

To accomplish this, the Dialogue Flow Manager monitors the current stage of the investigation, maintains conversation continuity, and identifies the most appropriate dialogue constraints for the current interaction. Rather than modifying the user’s intent, these constraints regulate how the response should be generated by reinforcing Sherlock Holmes’ investigative behavior and limiting unnecessary deviations from the ongoing narrative. The selected dialogue constraints are incorporated into the structured prompt before it is forwarded to the Response Generation Layer.

5.1.5 Response Generation Layer

The Response Generation Layer is responsible for producing Sherlock Holmes’ dialogue based on the structured prompt prepared by the middleware. Unlike the preceding components, this layer does not perform interaction management or evaluation. Instead, its sole responsibility is to generate a natural language response using the contextual information provided by the middleware.

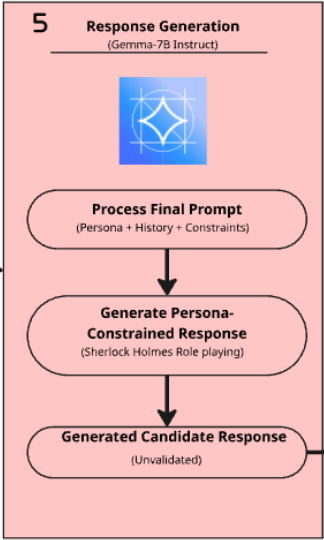


Figure 5.6: Response Generation Layer

Once the Dialogue Flow Manager completes the structured prompt, it is submitted to Gemma-7B Instruct, which serves as the response generation model of the proposed framework. The prompt contains the validated user interaction, the Persona Knowledge Base, the current dialogue state, conversation history, investigation context, and the selected dialogue constraints. By supplying this information explicitly, the middleware reduces the dependence on the language model’s internal memory while providing consistent contextual guidance for response generation (Gemma Team et al., 2024).

Gemma-7B Instruct processes the structured prompt and generates a candidate response that reflects Sherlock Holmes’ reasoning, personality, and communication style. Since conversation management has already been performed by the preceding middleware components, the language model can focus primarily on generating natural and contextually appropriate dialogue rather than determining how the conversation should progress. The generated response is then forwarded to the Output Validation Engine for final evaluation before it is presented to the user.

5.1.6 Output Validation Engine

The Output Validation Engine is the final processing component of the proposed Interaction-Layer Middleware. Its primary responsibility is to verify that the candidate response generated by the Response Generation Layer satisfies the persona and narrative requirements of the Sherlock Holmes role-playing environment before it is presented to the user. Rather than immediately delivering the first generated response, the proposed framework performs an independent evaluation to determine whether the response meets the required quality standards (Samuel et al., 2025; X. Wang et al., 2024; Wu, Sun, Bai, Li, & Wang, 2025).

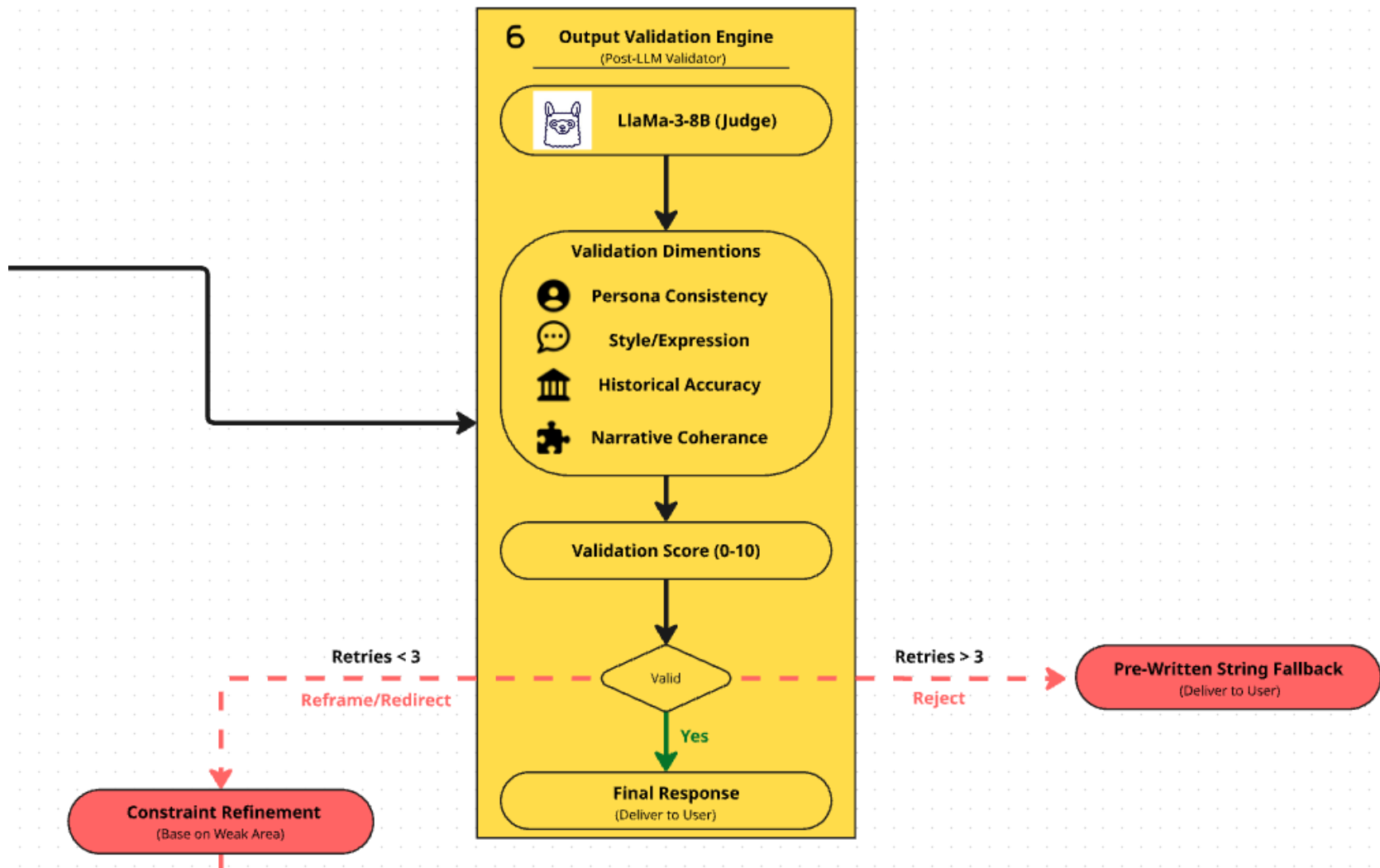


Figure 5.7: Output Validation Engine Architecture

After Gemma-7B Instruct generates a candidate response, it is evaluated by LLaMA-3-8B Instruct, which serves as the Judge LLM. The Judge assesses the response across four validation dimensions: Persona Consistency, Style and Expression, Historical Accuracy, and Narrative Coherence, and determines whether the response meets the framework’s validation requirements.

If the response passes validation, it is delivered to the user. Otherwise, the middleware identifies the weak evaluation dimensions and performs constraint refinement, where the corresponding dialogue constraints are strengthened before generating another response. This process may be repeated for a limited number of attempts to improve response quality while preserving the original intent of the conversation.

To prevent infinite regeneration loops, the framework maintains a retry counter. Once the maximum number of retries is reached, the regeneration process is terminated and a predefined fallback response is returned to the user. This ensures that every interaction receives a response while maintaining stable system behavior.

5.2 Interaction Evaluation Framework

The proposed Interaction-Layer Middleware regulates conversations using three evaluation mechanisms that operate at different stages of the interaction pipeline. Rather than relying solely on the response generation model to determine whether a conversation should continue, the middleware independently evaluates user interactions, dialogue progression, and generated responses before allowing the conversation to proceed. This layered evaluation approach enables the framework to maintain Sherlock Holmes’ persona while improving conversational consistency throughout extended interactions (Das & Fioretto, 2026; Samuel et al., 2025; Zhang et al., 2023).

Each evaluation mechanism is implemented using LLaMA-3-8B Instruct as the Judge LLM. Unlike Gemma-7B Instruct, which is responsible for generating Sherlock Holmes’ responses, the Judge LLM performs structured evaluations based on predefined rubrics. The evaluation results are then interpreted by the middleware to determine the appropriate action before the conversation advances to the next stage.

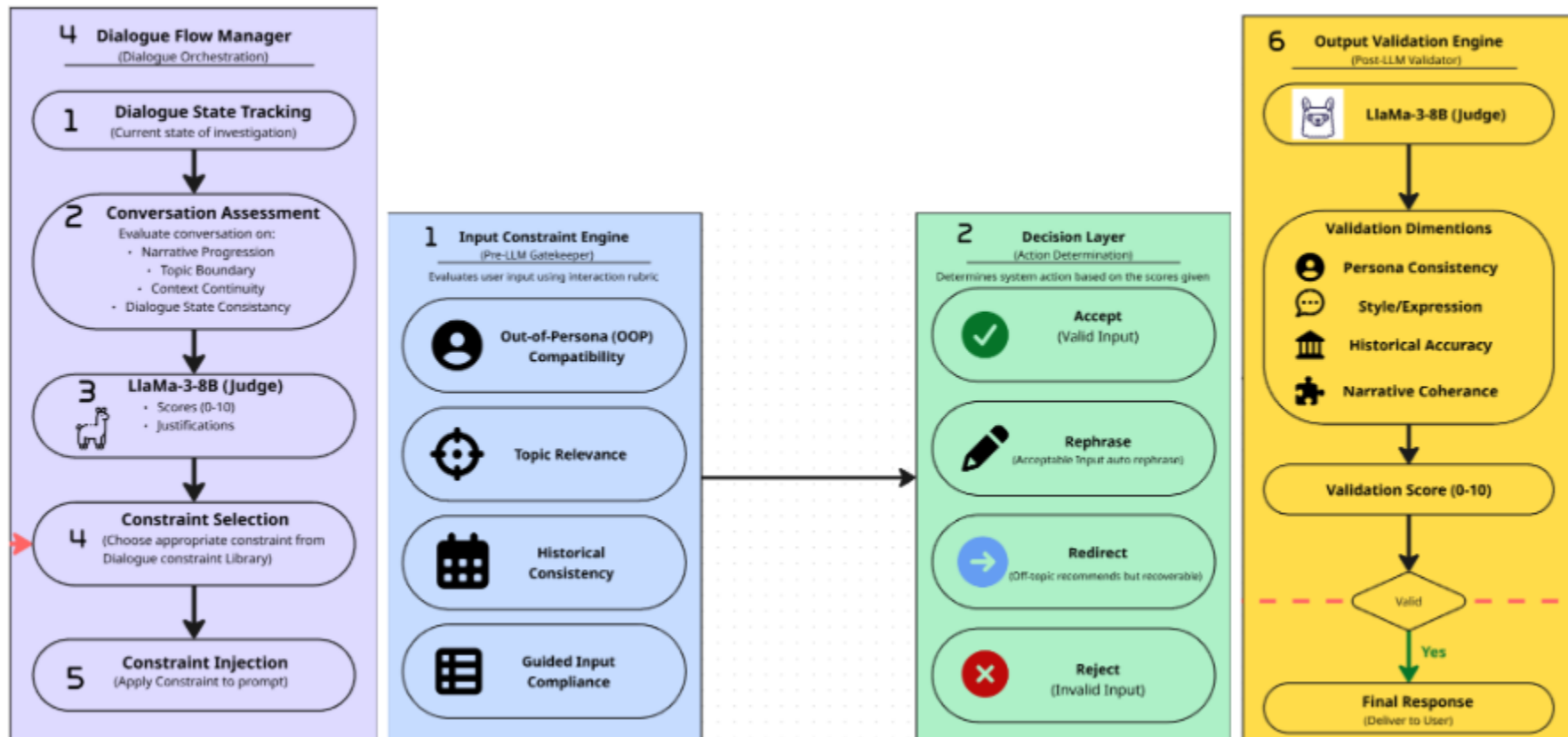


Figure 5.8: The Evaluation Stages

The framework consists of three evaluation stages:

- **Input Constraint Evaluation**, which evaluates user interactions before prompt construction.
- **Dialogue Flow Evaluation**, which monitors the progression of the conversation and selects appropriate dialogue constraints.
- **Output Validation Evaluation**, which evaluates the generated response before it is presented to the user.

Together, these evaluation mechanisms allow the middleware to regulate conversations before, during, and after response generation, reducing persona drift while maintaining coherent multi-turn role-playing interactions.

5.2.1 Input Constraint Evaluation

The Input Constraint Evaluation is performed immediately after the user submits a message to the chatbot. Its purpose is to determine whether the interaction is appropriate for the Sherlock Holmes role-playing environment before it reaches the Prompt Builder. By evaluating user interactions prior to response generation, the middleware minimizes interactions that may negatively affect persona consistency or disrupt the intended investigative narrative (Das & Fioretto, 2026; Kumar et al., 2026).

The Judge LLM evaluates each user interaction using four evaluation criteria.

Out-of-Persona Compatibility

Out-of-Persona Compatibility measures whether the user’s message can be answered while preserving Sherlock Holmes’ established identity. The evaluation determines whether the interaction encourages Sherlock Holmes to remain within his assigned role or attempts to override the intended character. Messages that support the role-playing environment receive higher scores, while interactions that encourage persona breaking receive lower scores.

Topic Relevance

Topic Relevance evaluates whether the user’s interaction contributes to the current investigation or discussion. Messages related to the active case, previously discussed evidence, or ongoing deductions receive higher scores because they

maintain the continuity of the conversation. Unrelated topics receive lower scores since they interrupt the progression of the investigation.

Historical Consistency

Historical Consistency evaluates whether the user’s message remains appropriate within the Victorian setting of Sherlock Holmes. Interactions that require knowledge beyond Holmes’ historical period receive lower scores, while requests that can be answered using information available during the nineteenth century receive higher scores.

Guided Input Compliance

Guided Input Compliance measures whether the user’s interaction follows the intended use of the chatbot. Clear investigative questions and logically structured requests receive higher scores, whereas ambiguous or incomplete interactions receive lower scores because they are more likely to produce inconsistent responses.

The four criterion scores are combined to produce the **Weighted Interaction Constraint Score (WICS)**:

$$\text{WICS} = (0.35 \times \text{OOP}) + (0.25 \times \text{TR}) + (0.25 \times \text{HC}) + (0.15 \times \text{GIP}) \quad (5.1)$$

where:

OOP = Out-of-Persona Compatibility

TR = Topic Relevance

HC = Historical Consistency

GIP = Guided Input Compliance

Theoretical Basis for Weighted Interaction Constraint Score (ICE Rubric)

The specific weight distributions applied to the Weighted Interaction Constraint Score rubric are established as preliminary heuristic values. Because no universally standardized weighting currently exists for evaluating user-driven inputs in role-playing LLM frameworks, these initial weights are derived directly from Persona Theory (Section 3.1) and Interaction Steering Theory (Section 3.3).

Table 5.1: Input Constraint weighted scoring and decision

Criterion	Weight	Justification
Out-of-Persona (OOP) Compatibility	35%	Assigned the highest weight because preserving the character’s core identity is the foundational objective of the proposed framework, grounded in the multi-layered Persona Object Model (Section 3.1). If a user interaction successfully forces the character outside of his established psychological and behavioral baseline, the role-playing experience is immediately compromised, rendering performance in the remaining criteria irrelevant.
Topic Relevance	25%	Assigned a high secondary weight based on Dialogue Policies and Narrative Progression (Section 3.2.3). This ensures the user’s input remains focused on the active investigation. Maintaining discussion within the narrative boundary prevents unnecessary topic drift, forcing the LLM to sustain coherent multi-turn interactions rather than hallucinating unrelated conversational branches.
Historical Consistency	25%	Weighted equally to Topic Relevance, this criterion is directly anchored in the Knowledge Layer’s strict Knowledge Boundaries (Section 3.1.3). It preserves the authenticity of the Victorian setting by explicitly preventing user interactions that introduce modern knowledge, technology, or concepts unavailable during Sherlock Holmes’ era. Breaching this boundary significantly reduces immersion and triggers temporal hallucinations in the LLM.
Guided Input Compliance	15%	Assigned the lowest weight. While this criterion improves the clarity of user interactions by encouraging well-formed investigative queries—aligning with Constraint-Based Interaction Steering (Section 3.3.2)—it is fundamentally a structural formatting check. Inputs that score poorly under this criterion can often be corrected through automatic rephrasing in the middleware without substantially threatening the LLM’s persona consistency.

Experimental Evaluation of Constraint Weight Variance on Output Validation

Because the WICS weights are preliminary and theory-driven, this study includes a systematic experiment to evaluate how varying these input-side weights affects the system’s downstream Output Validation task. The objective is to determine how strictness at the input constraint level (the ICE layer) influences the Output Validation Module’s ability to accurately catch or reject persona drift during the final LLM-as-a-Judge evaluation (Section 3.5.1). The experiment evaluates the pipeline across three distinct weight configurations for the ICE rubric to observe their downstream impact:

1. Baseline Configuration (Equal Weighting): All four input criteria are assigned an equal mathematical weight (25%). This acts as a control baseline to determine if prioritizing identity (OOP Compatibility) at the input stage actually improves downstream fidelity, or if a flat average is sufficient to steer the LLM.

2. Theory-Driven Configuration (Proposed): The preliminary theoretical weights (OOP Compatibility at 35%, Topic Relevance at 25%, Historical Consistency at 25%, Guided Input at 15%).

3. Ablation Configuration (Inverted Variance): The input pipeline is tested by shifting the primary weight away from core identity constraints and heavily skewing it toward structural formatting (e.g., assigning a 50% weight to Guided Input Compliance). This isolates the constraint mechanisms to observe how disproportionate weighting at the input stage forces the downstream validation system into failure states.

By processing the user interactions through these three input configurations, the experiment will measure how the variance impacts the final Judge LLM’s scoring of the generated responses. Specifically, the evaluation will monitor whether weakening the OOP Compatibility weight at the input stage increases the False Negative Rate at the output stage (where the validation module fails to detect an out-of-character response). The configuration that minimizes downstream validation errors and aligns most closely with human annotator benchmarks will establish the empirically optimal weight distribution for the interaction constraints.

After computing the WICS, the middleware compares the resulting score against predefined decision thresholds to determine the appropriate system action. Depending on the evaluation, the interaction may be accepted, automatically rephrased, redirected toward a more appropriate conversation, or rejected before prompt construction begins.

5.2.1.1 Decision Layer

After the Weighted Interaction Constraint Score (WICS) has been computed, the result is forwarded to the Decision Layer, which determines how the middleware should process the user’s interaction. Rather than allowing the Judge LLM to directly decide the system response, the Decision Layer interprets the evaluation scores using predefined decision thresholds. This separation allows the evaluation process and the decision-making process to remain independent while ensuring that every interaction is handled consistently.

The Decision Layer considers both the overall WICS and the individual criterion scores before selecting one of four possible actions: Accept, Automatic Rephrase, Redirect, or Reject. This prevents interactions with a high overall score but a critically low score in one criterion from bypassing important safeguards.

Accept

Interactions that satisfy the acceptance threshold are considered appropriate for the Sherlock Holmes role-playing environment. These interactions demonstrate strong persona compatibility, remain relevant to the current investigation, preserve the historical setting, and comply with the expected interaction format. Accepted interactions are forwarded directly to the Prompt Builder for structured prompt construction.

Rephrase

Interactions that are generally acceptable but contain minor issues are automatically rephrased before continuing through the middleware. Automatic rephrasing is primarily intended to improve the clarity or structure of the user’s request without changing its original meaning. This enables the system to recover interactions that may otherwise produce inconsistent responses.

Redirect

Interactions that deviate from the current investigation but remain recoverable are redirected toward a more appropriate conversational context. Rather than rejecting the user’s request outright, the middleware encourages the user to continue the conversation in a manner that better aligns with Sherlock Holmes’ investigative role.

Reject

Interactions that receive very low evaluation scores or violate critical evalua-

tion criteria are rejected before prompt construction. These interactions typically attempt to force Sherlock Holmes outside his established persona, introduce significant historical inconsistencies, or contain content that cannot be incorporated into the intended role-playing environment. Instead of forwarding these interactions to the response generation model, the middleware returns a predefined system message explaining that the request cannot be processed.

Table 5.2: WICS Decision Table

Decision	Weighted Score	Additional Rule	Description
Accept	$\text{WICS} \geq 8.0$	No criterion below 7	Input fully supports Sherlock Holmes’ persona and investigation.
Rephrase	$6.0 \leq \text{WICS} < 8.0$	One criterion between 5–6, none below 5	Input is acceptable and will be automatically rephrased.
Redirect	$4.0 \leq \text{WICS} < 6.0$	One criterion between 3–4, none below 3	Input is outside the current investigation but can be redirected to a supported interaction.
Reject	$\text{WICS} < 4.0$	Any criterion ≤ 2	Input violates one or more core interaction constraints and should not proceed to response generation.

5.2.2 Dialogue Flow Evaluation

Once a user interaction has been accepted and incorporated into the structured prompt, the conversation proceeds to the Dialogue Flow Manager. Unlike the Input Constraint Evaluation, which determines whether an interaction should enter the role-playing environment, the Dialogue Flow Evaluation regulates how the conversation should progress after the interaction has been accepted. Its primary objective is to maintain the continuity of the investigation by ensuring that responses remain logically connected to previous dialogue while preserving Sherlock Holmes’ reasoning process (Zhang et al., 2023).

The current conversation is evaluated using four criteria that collectively determine whether the investigation is progressing in a coherent and consistent manner.

Conversation State Tracking

Conversation State Tracking determines whether the dialogue remains within the appropriate stage of the investigation. The evaluation monitors transitions between stages such as case introduction, evidence gathering, suspect analysis, deduction, and case resolution. Maintaining the correct dialogue state prevents the conversation from advancing too quickly or revisiting completed stages without justification.

Narrative Progression

Narrative Progression evaluates how effectively the conversation develops toward solving the investigation. Rather than revealing conclusions immediately, responses should gradually introduce observations, evidence, and deductions that contribute to the overall narrative. This criterion encourages a natural investigative process that resembles Sherlock Holmes' method of reasoning.

Context Continuity

Context Continuity measures whether previously established information is preserved throughout the conversation. Responses should reference earlier evidence, suspects, observations, and deductions whenever appropriate while avoiding contradictions or unnecessary repetition. Maintaining continuity enables the investigation to develop as a connected dialogue rather than as isolated conversational turns.

Topic Boundary Control

Topic Boundary Control evaluates whether the conversation remains focused on the current investigation. Minor conversational deviations may occur naturally; however, significant topic shifts that interrupt the investigation reduce the overall evaluation score. This criterion helps maintain narrative focus while allowing the conversation to remain flexible.

The four criterion scores are combined to produce the **Dialogue Flow Management Score (DFMS)**:

$$\text{DFMS} = (0.35 \times \text{CST}) + (0.30 \times \text{NP}) + (0.20 \times \text{CC}) + (0.15 \times \text{TBC}) \quad (5.2)$$

where:

CST = Conversation State Tracking

NP = Narrative Progression

CC = Context Continuity
TBC = Topic Boundary Control

Theoretical Basis for Dialogue Flow Management Score (DFMS) Weights

The specific weight distributions applied to the Dialogue Flow Management Score (DFMS) are established as preliminary heuristic values. Because no standardized mathematical weighting currently exists for evaluating interaction-constrained LLMs in role-playing contexts, these initial weights are derived directly from Conversation Management Theory (Section 3.2) and the hierarchical dependencies of the proposed architecture.

Table 5.3: Dialogue Flow weighted scoring and decision criteria

Criterion	Weight	Justification
Conversation State Tracking	35%	Assigned the highest weight, grounded in Dialogue State Tracking theory (Section 3.2.1). Theoretical models indicate that external state management is the foundational anchor for reducing inconsistencies (Khan et al., 2025; Zhang et al., 2023). If the system loses track of the conversational stage, every downstream judgment becomes inherently unreliable, leading to systemic failure.
Narrative Progression Rules	30%	Weighted second-highest, based on the principles of Dialogue Policies and Narrative Progression (Section 3.2.3). Regulating conversational flow toward predefined objectives is the most critical immersion risk once the state is established. Prematurely revealing a culprit or deduction violates the dialogue policy and creates a permanent narrative break that, unlike a stylistic error, cannot be recovered.
Context Continuity	20%	Assigned a moderate weight, derived from Conversation Continuity principles (Section 3.2.2). While failing to connect responses to previous exchanges negatively impacts logical consistency, it is scoped lower than State Tracking. Dropped contextual threads degrade the user experience but are fundamentally more recoverable than a broken conversational stage.
Topic Boundary Control	15%	Assigned the lowest weight, mapping to Constraint-Based Interaction Steering (Section 3.3.2). While regulating topic drift is necessary to prevent deviation from the intended character, it acts as a peripheral boundary constraint. Topic drift is a localized error that is easily corrected with a conversational redirect, making it the least structurally damaging and most recoverable deviation.

Experimental Evaluation of Weight Variance on Output Validation

Since the initial DFMS weights are preliminary and theory-driven, this study includes a systematic experiment to evaluate how varying these weight distributions directly affects the system’s output validation task. The objective is to determine how shifts in the Dialogue Flow constraints alter the Output Validation Module’s ability to accurately catch or reject persona drift during the LLM-as-a-Judge evaluation stage (Section 3.5.1).

The experiment evaluates the generated dialogue dataset across three distinct weight configurations to observe their impact on the validation threshold:

1. Baseline Configuration (Equal Weighting): All four criteria are assigned equal mathematical weights (25%). This establishes a control baseline to test if the theoretical hierarchy improves the validation module’s accuracy, or if a flat average is sufficient for the LLM judge to detect drift.

2. Theory-Driven Configuration (Proposed): The preliminary theoretical weights (CST at 35%, NPR at 30%, CC at 20%, TBC at 15%).

3. Ablation Configuration (Inverted Variance): The pipeline is tested by heavily skewing weights toward the lower-priority criteria (e.g., assigning a 60% weight to Topic Boundary Control). This isolates specific constraints to observe how disproportionate weighting forces the validation system into failure states.

By passing the outputs of these three configurations through the Response Validation pipeline, the experiment will measure how weight variance impacts the final LLM Judge’s scoring. The configuration will be evaluated based on its alignment with human annotators, specifically monitoring how varying the weights increases or decreases the **False Positive Rate** (the validation module incorrectly rejecting a valid, character-consistent response) and the **False Negative Rate** (the validation module accepting a response containing critical narrative or state-tracking errors). The configuration that minimizes these validation errors will determine the optimal weight distribution for the framework.

After computing the DFMS, the Dialogue Flow Manager determines whether additional conversational guidance is required before response generation. Instead of directly modifying the user’s message, the framework selects appropriate dialogue constraints from the Dialogue Constraint Library based on the evaluation results. These constraints are incorporated into the structured prompt to guide the succeeding response while preserving the intended narrative and Sherlock Holmes’ investigative behavior.

Dialogue Constraint Library

The Dialogue Constraint Library contains a collection of predefined constraints that guide the response generation process according to the current conversational context. Rather than generating new constraints dynamically, the framework selects the most appropriate constraints based on the Dialogue Flow Evaluation and incorporates them into the structured prompt before it is forwarded to the Response Generation Layer.

The constraint library is organized into six categories:

- **Dialogue State Constraints**, which maintain the current stage of the investigation.
- **Narrative Progression Constraints**, which regulate how observations, evidence, and deductions are introduced throughout the conversation.
- **Topic Boundary Constraints**, which maintain the discussion within the scope of the active investigation.
- **Context Continuity Constraints**, which reinforce previously established information and reduce contradictions.
- **Persona Constraints**, which preserve Sherlock Holmes’ analytical reasoning, personality, and communication style.
- **Historical Constraints**, which prevent references that conflict with the Victorian setting of the role-playing environment.

Depending on the current dialogue state, multiple constraint categories may be applied simultaneously. The selected constraints are combined with the structured prompt produced by the Prompt Builder to create the final prompt submitted to the Response Generation Layer. By regulating the conversation through predefined constraints rather than modifying the language model itself, the proposed framework maintains greater control over dialogue progression while preserving the flexibility of the underlying response generation model.

5.2.3 Output Validation Evaluation

The final stage of the proposed Interaction Evaluation Framework is the Output Validation Evaluation, which verifies the quality of the candidate response before

it is delivered to the user. While the previous evaluation stages regulate user interactions and dialogue progression, the Output Validation Engine focuses on the generated response itself. Its objective is to ensure that every response remains faithful to Sherlock Holmes’ established persona, supports the current investigation, and maintains the overall quality of the conversation before reaching the user (Samuel et al., 2025; X. Wang et al., 2024; Wu et al., 2025).

The candidate response generated by Gemma-7B Instruct is evaluated using four validation criteria that collectively measure the overall quality of the response.

Persona Consistency

Persona Consistency evaluates whether the generated response accurately reflects Sherlock Holmes’ established personality, reasoning process, and behavioral characteristics. Responses should demonstrate Holmes’ analytical thinking, logical deductions, and reserved manner of communication. Responses that contradict his established identity or exhibit behaviors inconsistent with the character receive lower evaluation scores.

Narrative Coherence

Narrative Coherence measures whether the response logically follows the current investigation and remains consistent with the established dialogue. The response should contribute meaningfully to the progression of the case without introducing contradictions, abrupt topic changes, or unsupported conclusions.

Historical and World Boundary

Historical and World Boundary evaluates whether the generated response remains appropriate within the Victorian setting of Sherlock Holmes. References to modern technologies, contemporary events, or knowledge unavailable during the historical period reduce the evaluation score because they disrupt the authenticity of the role-playing environment.

Style and Syntax

Style and Syntax evaluates whether the language used in the response resembles Sherlock Holmes’ characteristic communication style. The evaluation considers sentence structure, vocabulary, tone, and overall writing style to determine whether the response maintains a consistent literary presentation throughout the conversation.

The individual criterion scores are combined to compute the **Output Validation Score (OVS)**:

$$\text{OVS} = (0.40 \times \text{PCC}) + (0.30 \times \text{NC}) + (0.20 \times \text{HWB}) + (0.10 \times \text{SS}) \quad (5.3)$$

where:

PCC = Persona Consistency

NC = Narrative Coherence

HWB = Historical and World Boundary

SS = Style and Syntax

The weighting scheme reflects the priorities of the proposed framework. Persona Consistency receives the highest weight because preserving Sherlock Holmes’ identity is the primary objective of the chatbot. Narrative Coherence is assigned the second-highest weight since responses must remain logically connected to the ongoing investigation. Historical and World Boundary ensures that responses remain faithful to the intended setting, while Style and Syntax contributes to maintaining a consistent conversational style throughout the interaction.

After computing the OVS, the middleware compares the resulting score against the predefined validation thresholds to determine the appropriate system action. Responses that satisfy the required quality threshold are immediately presented to the user. Responses that receive lower scores undergo additional processing through the Constraint Refinement Module, where the Dialogue Constraint Library is updated by strengthening the constraints associated with the weak evaluation dimensions before another response is generated.

To prevent repeated regeneration of low-quality responses, the framework maintains a retry counter for each interaction. Once the maximum retry limit has been reached, the regeneration process is terminated and a predefined fallback response is returned to the user. This mechanism ensures that every interaction receives a response while preventing unnecessary computational overhead caused by continuous regeneration attempts.

Theoretical Basis for Output Validation Score (OVS) Weights

The specific weight distributions applied to the Output Validation Score (OVS) are established as preliminary heuristic values. Because no standardized mathematical weighting currently exists for evaluating final role-playing responses using LLM-as-a-Judge frameworks, these initial weights are derived directly from the Persona Object Model (Section 3.1), Conversation Management Theory (Section 3.2), and the proposed Interaction-Layer Middleware architecture. The assigned weights reflect the theoretical priority of each validation dimension in preserving Sherlock Holmes’ persona while maintaining coherent and historically consistent

multi-turn conversations.

Table 5.4: Output Validation weighted scoring and decision criteria

Evaluation Metric	Weight	Justification
Persona Consistency	40%	Assigned the highest weight because preserving Sherlock Holmes’ identity is the primary objective of the proposed framework, directly grounded in the Persona Object Model (Section 3.1). If the generated response contradicts Holmes’ established personality, reasoning style, or behavioral characteristics, the role-playing illusion immediately collapses regardless of the response’s grammatical quality or logical structure. Consequently, Persona Consistency serves as the primary indicator of overall response quality.
Narrative Coherence	30%	Assigned the second-highest weight based on Conversation Management Theory and Narrative Progression (Section 3.2). Even if the response remains in character, it must logically follow the ongoing investigation, preserve dialogue continuity, and contribute meaningfully to the progression of the case. Responses containing contradictions, unsupported deductions, or abrupt topic transitions significantly disrupt the investigative narrative and reduce conversation quality.
Historical/World Boundary	20%	Assigned a moderate weight because it preserves the authenticity of Sherlock Holmes’ Victorian setting through the Knowledge Layer’s established knowledge boundaries (Section 3.1.3). References to modern technologies, contemporary events, or knowledge unavailable during Holmes’ era reduce immersion and introduce temporal inconsistencies. However, these violations are generally less damaging than breaking the character’s personality or the logical progression of the investigation.

Continued on next page

Evaluation Metric	Weight	Justification
Style & Syntax	10%	Assigned the lowest weight because stylistic variations are generally more recoverable than errors involving persona, narrative, or historical accuracy. While maintaining Holmes’ characteristic vocabulary, sentence structure, and tone contributes to immersion, minor stylistic inconsistencies rarely prevent users from recognizing the intended character. Therefore, Style and Syntax functions primarily as a refinement criterion rather than a core validation requirement.

Experimental Evaluation of Output Validation Weight Variance

Because the initial OVS weights are preliminary and theory-driven, this study includes a systematic experiment to evaluate how varying these output-side weight distributions affects the framework’s response validation performance. The objective is to determine whether the proposed theoretical hierarchy enables the Judge LLM to more accurately distinguish high-quality persona-consistent responses from responses containing persona drift or narrative inconsistencies.

The experiment evaluates the Output Validation Engine across three distinct weight configurations:

1. Baseline Configuration (Equal Weighting)

All four validation criteria are assigned equal mathematical weights (25%). This serves as a control configuration to determine whether prioritizing particular validation dimensions improves response evaluation or whether an equal contribution from all criteria is sufficient.

2. Theory-Driven Configuration (Proposed)

The preliminary theoretical weights are applied:

- Persona Consistency – 40%
- Narrative Coherence – 30%
- Historical and World Boundary – 20%
- Style and Syntax – 10%

3. Ablation Configuration (Inverted Variance)

The weighting priority is intentionally shifted toward lower-priority stylistic evaluation (e.g., assigning a 50–60% weight to Style and Syntax while reducing Persona Consistency). This configuration isolates the effect of disproportionate weighting and evaluates whether emphasizing superficial linguistic quality reduces the Judge LLM’s ability to identify genuine persona inconsistencies. By evaluating the same generated responses across these three weighting configurations, the study measures how weight variation influences the final validation decisions of the Output Validation Engine. The configurations will be compared using their agreement with human evaluators while monitoring changes in the False Positive Rate (incorrectly rejecting a valid response) and False Negative Rate (incorrectly accepting a response containing persona or narrative violations). The configuration that minimizes these validation errors while achieving the highest agreement with human judgments will be identified as the empirically optimal weight distribution for the Output Validation Score.

Appendix A

Research Ethics Documents



RESEARCH / PROJECT ETHICS REVIEW FORM
For Thesis / Capstone / Dissertation Proposals
(ver.Oct2024c)

Names of Student Researcher(s):

Van Asher Alcantara

Jorien Encarguez II

Austin Philippe Gan

Allen Andrei Gutierrez

Research/Thesis/Capstone Title: Curbing the Drift: How Regulated Conversational Flows Shape LLM Persona Consistency

College: College of Computer Studies

Department: Department of Software Technology

Course: CS-ST

Expected Duration of the Project: from: May 4, 2026 to Apr 6, 2027

PROTOCOL:

(To be filled up by proponents; Use the Ethics Checklists as guides in determining areas for ethical concern/consideration. Attach and properly reference the accomplished ethics checklists and supporting documents.)

		Reference Document (include page number)
Target participants	Undergraduate participants aged 18 years or older who possess sufficient familiarity with Sherlock Holmes' personality, communication style, and investigative behavior to recognize deviations in the character's persona during conversation.	1.3 Scope and Limitations (pp. 4–6) 1.4 Methodology (pp. 6–10)
Data to be used / collected	- Multi-turn conversation logs generated during participant interactions - Participant evaluation data from questionnaires assessing the System Usability Scale (SUS)	1.4 Methodology (pp. 6–10)
Brief procedure	Participants evaluate four chatbot configurations using a within-subjects experimental design. In each session, participants engage in a multi-turn Sherlock Holmes role-playing conversation based on the same or equivalent fictional investigative scenario. After each session, participants complete evaluation questionnaires, and the collected conversation logs and questionnaire responses are analyzed together with automated	1.4 Methodology (pp. 6–10)

	evaluation metrics to assess persona consistency, usability, and responsiveness.	
Potential risks	The study presents minimal risk. Participants interact only with an AI chatbot portraying a fictional literary character. No sensitive personal information or psychologically distressing topics are intentionally introduced. Participation is voluntary, and participants may withdraw at any time without penalty.	1.3 Scope and Limitations (pp. 4–6) 1.4 Methodology (pp. 6–10)
Applicable Terms of Use or Licensing Agreement of Data, Models and/or Tools	The study uses publicly available, public-domain Sherlock Holmes literary works that are properly cited and used solely for academic, non-commercial research. The language models, software libraries, and development tools are used in accordance with their respective licenses and terms of use.	1.3 Scope and Limitations (pp. 3–6) and Section 1.4 Methodology (pp. 6–7)
Steps to safeguard the participants and/or data	• Participants provide informed consent before participation. • Participants may withdraw from the study at any time without penalty. • No personally identifiable information beyond what is necessary for participation will be collected. • Conversation logs are anonymized for analysis. • All research data are stored in password-protected storage accessible only to the researchers. • The chatbot discloses that it is an artificial intelligence system portraying the fictional character Sherlock Holmes.	1.4 Methodology (pp. 6–10)
Other concerns and considerations	Collected data will be stored for three (3) years and will be used in preparing technical reports and papers for conference proceedings and/or journals.	

PANEL'S ASSESSMENT/COMMENTS:

PANEL'S DECISION:

- ☒ **Approved** (*Proponent/s may proceed with the study/data collection*)
- ☐ **Modifications Required** (*Proponent/s should address ethical issues, follow the panel's recommendations for required modifications, and resubmit Ethics Review Form for approval before proceeding with the study or before starting with data collection*)
- ☐ **For Full Review** (*Proponent/s should submit the proposal to the University Research Ethics Review Committee (RERC) and obtain a research ethics clearance before proceeding with the study or before starting with data collection*)

JUSTIFICATION:


.....



Name and Signature of Adviser/Mentor

Date:




.....

Kristine Kalaw


.....

Maria Art Clarino

Name and Signature of Panel Member

Date:



Name and Signature of Panel Member

Date:




.....

Ethel Ong

Name and Signature of Panel Member

Date:



Appendix B

Turnitin Report

PAPER NAME

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AUTHOR

-

WORD COUNT

18890 Words

CHARACTER COUNT

124120 Characters

PAGE COUNT

61 Pages

FILE SIZE

1.7MB

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EXCLUDED OVERLAPPING SOURCES

Submitted to De La Salle University - Manila on 2026-07-13

77%

Submitted works

Appendix C

Detailed Declaration of AI Usage

This appendix provides a detailed record of all substantial uses of Generative Artificial Intelligence (AI) during the preparation of this thesis proposal. All AI-assisted outputs were reviewed, verified, revised where necessary, and approved by the authors before inclusion in the thesis. Generative AI tools were used only to assist with scaffolding, organization, language refinement, and improving the clarity of written content. No AI-generated content was accepted without critical review by the authors.

Table C.1: Detailed Declaration of AI Usage

Section	AI Tool	Description of AI Use	Integration Level
Chapter 1 – Introduction	ChatGPT, Gemini	Assisted in organizing the background of the study, refining the statement of the problem, objectives, significance of the study, and improving grammar, sentence structure, and overall coherence.	S, R
Chapter 2 – Review of Related Literature	ChatGPT, Gemini	Assisted in organizing the literature into thematic sections, improving transitions between related studies, refining paragraph structure, paraphrasing for clarity, and improving academic writing while preserving the authors’ interpretation of the cited literature.	S, R

Continued on next page

Table C.1 – Continued from previous page

Section	AI Tool	Description of AI Use	Integration Level
Chapter 3 – Methodology	ChatGPT, Gemini	Assisted in refining descriptions of the research methodology, improving the presentation of research phases, evaluation procedures, and system development process. Also helped improve readability and consistency of terminology throughout the chapter.	S, R
Chapter 4 – Persona Knowledge Base Design	ChatGPT, Gemini	Assisted in organizing discussions on the Persona Object Model, refining explanations of personality, expression, knowledge, beliefs, and moral values, and improving clarity and flow of the chapter.	S, R
Chapter 5 – Interaction-Layer Middleware Design	ChatGPT, Gemini	Assisted in refining explanations of the Input Constraint Engine, Decision Layer, Prompt Builder, Dialogue Flow Manager, Response Generation Layer, and Output Validation Engine. Helped improve organization, readability, and consistency of technical descriptions.	S, R
References and Formatting	ChatGPT, Gemini	Assisted in checking APA formatting consistency, improving citation formatting, refining table layouts, and reviewing the document for grammar, consistency, and typographical errors.	R
Abstract	ChatGPT, Gemini	Assisted in improving sentence structure, clarity, grammar, and overall organization while preserving the authors' intended meaning.	R

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